

Linear wave equations on Schwarzschild de Sitter spacetimes

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Abstract

This thesis studies linear wave equations on Schwarzschild–de Sitter spacetimes using methods from spectral theory and microlocal analysis. The Schwarzschild–de Sitter geometry provides a mathematical model of a black hole spacetime with positive cosmological constant and serves as an important setting for investigating the long-time behavior of wave propagation in general relativity.

The primary objective of this work is to present a detailed exposition of several fundamental results that are often treated only briefly in the literature. Building upon the review article of Häfner [10], we develop the analytical framework underlying the study of wave equations on black hole backgrounds, including the geometry of Schwarzschild–de Sitter spacetime, the associated wave operator, spectral methods and basic tools of microlocal analysis. Particular attention is devoted to the role of trapping, resonances, and radial point estimates in establishing decay properties of solutions.

In addition to surveying existing results, the thesis contains several detailed calculations and derivations that clarify intermediate steps omitted in the review literature. We reproduce key arguments and carry out independent computations that recover known results through the same underlying analytical techniques. These calculations are intended to provide a more transparent understanding of the methods rather than to establish new theoretical results.

The thesis demonstrates how spectral and microlocal techniques yield a powerful framework for studying linear wave equations on curved spacetimes. By presenting both the geometric background and the analytical tools in a unified manner, this work aims to serve as an accessible introduction to modern methods in the mathematical analysis of wave propagation on black hole spacetimes.

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Introduction

The study of wave propagation on curved spacetimes occupies a central position in mathematical general relativity. Linear wave equations arise naturally as simplified models for gravitational perturbations, electromagnetic fields, and quantum fields propagating on a fixed gravitational background. Understanding the long-time behaviour of their solutions is an essential step toward the analysis of stability problems for black hole spacetimes.

Among the simplest non-trivial black hole geometries is the Schwarzschild–de Sitter spacetime, which describes a spherically symmetric black hole in the presence of a positive cosmological constant. The geometry possesses both an event horizon and a cosmological horizon and therefore provides a natural setting in which to study scattering, decay, and resonance phenomena for wave equations.

Over the last several decades, spectral and microlocal methods have become fundamental tools in the analysis of wave equations on black hole backgrounds. Early investigations of scattering theory on black hole spacetimes were initiated by Dimock and Kay [7] and later developed by Bachelot [1]. These works established rigorous frameworks for studying the asymptotic behaviour of solutions and ultimately contributed to mathematical formulations of the Hawking effect.

A major development was the realization that decay properties of solutions are closely connected to analytic properties of the stationary resolvent obtained after Fourier transformation in time. This connection led to the study of resonances, which may be viewed as poles of meromorphic continuations of the resolvent. Important contributions in this direction include the resonance analysis of Schwarzschild–de Sitter spacetimes by Sá Barreto and Zworski [2] and the resonance expansions obtained by Bony and Häfner [3]. Later work of Melrose, Sá Barreto and Vasy [11], Dyatlov [8], and Vasy [13] developed a robust microlocal framework allowing the study of wave equations on a large class of black hole spacetimes.

There is also a complementary physical-space approach to the analysis of wave equations on black hole spacetimes, based on the vector-field method. In this framework, decay and boundedness are obtained by using energy currents, multiplier estimates, commutation vector fields, red-shift estimates near the horizons, and Morawetz-type integrated local energy decay estimates. In the Schwarzschild–de Sitter setting, Dafermos and Rodnianski proved decay results for solutions of the linear wave equation using this method [4]. Their lecture notes [5] give a broader exposition of the vector-field method for black hole spacetimes.

The present thesis is based primarily on the review article of Häfner [10], which surveys several aspects of spectral and microlocal analysis in mathematical general relativity. Our goal is not to obtain new results but rather to provide a detailed exposition of some of the arguments that are only sketched in the review. Particular emphasis is placed on the Schwarzschild–de Sitter model, the construction of Regge–Wheeler coordinates, the Fourier transformed wave operator, Fredholm theory, and the role of Sobolev regularity near the horizons.

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1. Framework

1.1. Penrose Diagram and the Schwarzschild de Sitter Metric

We begin with the Schwarzschild de Sitter metric in Boyer–Lindquist coordinates for the manifold $M = \mathbb{R}_t \times \tilde{X}$ where $\tilde{X} = \mathcal{I}_{b_0} \times S_\omega^2$, $b_0 = (m, 0)$ and $\mathcal{I}_{b_0} = (r_-, r_+)$ by

$$g = F_{b_0}(r) dt^2 - F_{b_0}(r)^{-1} dr^2 - r^2 d\omega^2, \quad F_{b_0}(r) = 1 - \frac{2m}{r} - \frac{\Lambda r^2}{3},$$

where $m > 0$ is the mass of the black hole, $\Lambda > 0$ is the cosmological constant, and $d\omega^2$ is the round metric on $\mathbb{S}^2$. Under the condition $9m^2\Lambda < 1$, the function F has two positive roots, denoted by r_- and r_+ . The region

$$r_- < r < r_+$$

is the static exterior region between the black-hole horizon and the cosmological horizon. The hypersurface $r = r_-$ is the future event horizon, while $r = r_+$ is the future cosmological horizon.

The coordinates (t, r, ω) are singular at the horizons because $F(r_\pm) = 0$. This is a coordinate singularity rather than a geometric singularity. To obtain coordinates which extend smoothly across the horizons, one introduces the Regge–Wheeler coordinate r_* , defined by

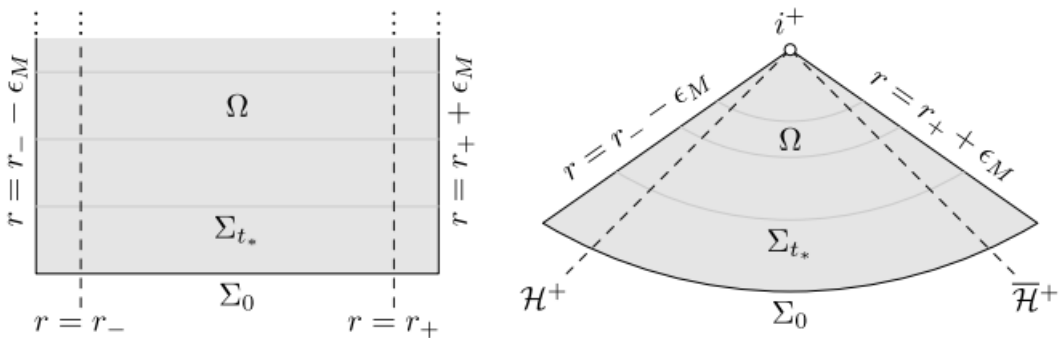
$$\frac{dr_*}{dr} = F(r)^{-1}.$$

One then replaces the Boyer–Lindquist time coordinate t by a modified time coordinate t_* . Near the black-hole horizon, t_* behaves like $t + r_*$, while near the cosmological horizon it behaves like $t - r_*$. This choice removes the coordinate degeneracy at the horizons and allows the metric to be extended to a slightly larger radial interval

$$X = (r_- - \epsilon_M, r_+ + \epsilon_M) \times \mathbb{S}^2,$$

for some small $\epsilon_M > 0$. Hence the extended spacetime region is written as

$$M = \mathbb{R}_{t_*} \times X.$$



Häfner [10][Figure 2.1]

Figure 1: The choice of coordinates

The purpose of Figure 1 is to give a compact representation of the causal and coordinate structure of the exterior region of a De Sitter–Schwarzschild black hole. Although the diagram is two-dimensional, it represents a four-dimensional spherically symmetric spacetime. The reduction to two dimensions comes from quotienting by the rotational symmetry group $SO(3)$. More precisely, since the angular variable $\omega \in \mathbb{S}^2$ is acted on by $SO(3)$, we consider the space

$$Q = M/SO(3)$$

Each point of Q represents an entire two-sphere in the original spacetime. Thus the Penrose diagram does not show individual spacetime points, but rather the geometry of the spacetime, with the angular directions suppressed.

In Figure 1, the dashed hypersurfaces $r = r_-$ and $r = r_+$ are null hypersurfaces. They correspond respectively to the future event horizon

$$\mathcal{H}^+ = \mathbb{R}_{t_*} \times \{r_-\} \times \mathbb{S}^2$$

and the future cosmological horizon

$$\overline{\mathcal{H}}^+ = \mathbb{R}_{t_*} \times \{r_+\} \times \mathbb{S}^2.$$

The parameter ϵ_M simply indicates that we have extended the coordinate patch slightly beyond these two horizons. The hypersurface Σ_0 is the initial spacelike hypersurface, and the hypersurfaces Σ_{t_*} are its translates under the flow of ∂_{t_*} . The shaded region

$$\Omega = \bigcup_{t_* \geq 0} \Sigma_{t_*}$$

is the future region swept out by this foliation. This is the region in which the evolution problem is studied.

The right-hand side of Figure 1 is a two-dimensional picture of the same spacetime region. Since we have quotiented out the angular $\mathbb{S}^2$ directions, each point in the diagram represents a whole two-sphere in the original four-dimensional spacetime. The diagram therefore only shows the behaviour of the time and radial directions.

The main idea of a Penrose diagram is to draw the spacetime in such a way that light rays are easy to see. In relativity, the paths followed by light are called null curves. In the diagram these are drawn at 45° . Curves which represent the motion of physical observers are timelike curves; these stay inside the light-cone directions. Thus the diagram shows the causal structure of the spacetime, meaning it shows which regions can send signals to which other regions. The point i^+ represents future timelike infinity: it is not an actual point of the spacetime, but an ideal endpoint for observers travelling into the far future.

The two diagonal boundary lines labelled $r = r_-$ and $r = r_+$ represent the two horizons. They are drawn as lightlike boundaries, meaning that they are generated by light rays. The surface $r = r_-$ is the black-hole horizon, while $r = r_+$ is the cosmological horizon. Σ_{t_*} denote the hypersurfaces. The family of slices Σ_{t_*} fills the region

$$\Omega = \bigcup_{t_* \geq 0} \Sigma_{t_*}.$$

Thus Ω is the future region obtained by evolving the initial hypersurface Σ_0 .

The importance of Figure 1 is that it shows that the coordinates (t_*, r, ω) remain well behaved when the slices Σ_{t_*} cross the horizons. This is why the coordinate t_* is used instead of the original Boyer–Lindquist time coordinate t . The original coordinate t becomes singular at $r = r_-$ and $r = r_+$, whereas the coordinate t_* allows us to describe the geometry smoothly across these hypersurfaces. This makes the diagram useful for setting up evolution problems, because one can study equations on the whole region Ω without the coordinate system breaking down at the horizons.

Using these coordinates,

$$g = \begin{cases} F dt_*^2 - 2 dt_* dr - r^2 d\omega^2 & \text{near } r = r_-, \\ F dt_*^2 + 2 dt_* dr - r^2 d\omega^2 & \text{near } r = r_+. \end{cases}$$

Now, using the definition of the Laplace Beltrami operator i.e.

$$\square_g u = \frac{1}{\sqrt{|g|}} \partial_\mu \left(\sqrt{|g|} g^{\mu\nu} \partial_\nu u \right) \quad (1.1.1)$$

since $t_* = t + r_*$ in a neighbourhood of the event horizon and $t_* = t - r_*$ in a neighbourhood of the cosmological horizon, the wave equation takes the following form near the corresponding horizons:

$$\square_g = \begin{cases} -2\partial_{t_*} \partial_r - F \partial_r^2 - \frac{2}{r} \partial_{t_*} - \left(F' + \frac{2F}{r}\right) \partial_r - r^{-2} \Delta_{\mathbb{S}^2} & \text{near } r = r_-, \\ 2\partial_{t_*} \partial_r - F \partial_r^2 + \frac{2}{r} \partial_{t_*} - \left(F' + \frac{2F}{r}\right) \partial_r - r^{-2} \Delta_{\mathbb{S}^2} & \text{near } r = r_+ \end{cases} \quad (1.1.2)$$

Now let

$$\kappa_\pm = \mp \frac{1}{2} F'(r_\pm)$$

and define

$$\gamma = \max\left\{\frac{1}{\kappa_-}, \frac{1}{\kappa_+}\right\}$$

Under this framework, we will prove the following theorem.

Theorem 1.1.1 ([10] Theorem 5.1). *Let $s \geq \frac{1}{2}$, $\mathcal{X}^s = \{u \in \bar{H}^s(X) : \hat{\square}_g(0)u \in \bar{H}^{s-1}(X)\}$ and $\mathcal{Y}^{s-1} = \bar{H}^{s-1}$. Then the operators*

$$\hat{\square}_g(\sigma) : \mathcal{X}^s \rightarrow \mathcal{Y}^{s-1}$$

form an analytic family of Fredholm operators of index 0 for all $\sigma \in \mathbb{C}$ such that

$$\text{Im}(\sigma) > \frac{1-2s}{2\gamma} \quad (1.1.3)$$

Moreover, $\hat{\square}_g(\sigma)$ is invertible if $\text{Im}(\sigma) > 0$.

1.2. Motivation for Regge–Wheeler Coordinates: Analysis in Boyer–Lindquist Coordinates

The purpose of this section is to understand the behaviour of mode solutions near r_- and r_+ in Boyer–Lindquist coordinates, thereby providing additional motivation for the introduction of the t_* coordinate. Since the coefficients of the Schwarzschild–de Sitter metric become singular in Boyer–Lindquist coordinates at these hypersurfaces, it is natural to ask how solutions of the Fourier transformed wave equation behave near the horizons. By separating variables and performing a local analysis of the resulting radial equation, we will see that the singular coordinate system produces oscillatory branches of the form $(r - r_h)^{\pm i\sigma/c}$. This observation motivates the introduction of Regge–Wheeler coordinates, in which one of the branch extends smoothly across the horizons and the Fredholm theory can be formulated in natural Sobolev spaces. Let us first try to solve the homogeneous wave equation without the use of Regge–Wheeler coordinates and see what kind of issues we face.

The wave operator is given by

$$\square_{g_{b_0}} = F_{b_0}^{-1} \partial_t^2 - \frac{1}{r^2} \partial_r r^2 F_{b_0} \partial_r - r^{-2} \Delta_{\mathbb{S}^2}$$

Let $P(\sigma) = F_{b_0} \tilde{\square}_{g_{b_0}}(\sigma)$ where $\tilde{\square}_{g_{b_0}}$ denotes the Fourier transform of the operator with respect to t .

Then

$$P(\sigma) = -\sigma^2 - \frac{F_{b_0}}{r^2} \partial_r (r^2 F_{b_0} \partial_r) - \frac{F_{b_0}}{r^2} \Delta_{\mathbb{S}^2}$$

We want to solve the homogeneous problem

$$P(\sigma)\hat{u} = 0 \quad (1.2.1)$$

using the Frobenius method and see why doing it in the Boyer Lindquist coordinates makes it look like a naive approach.

$$\hat{u}(\sigma, r, \theta, \phi) = R(r)Y_{lm}(\theta, \phi)$$

where $Y_{lm}(\theta, \phi)$ are the spherical harmonics that satisfy $\Delta_{\mathbb{S}^2}Y_{lm} = -l(l+1)Y_{lm}$. Then equation 1.2.1 becomes

$$-\sigma^2 R Y_{lm} - \frac{F_{b_0}}{r^2} \partial_r (r^2 F_{b_0} \partial_r R) Y_{lm} - \frac{F_{b_0}}{r^2} R (-l(l+1) Y_{lm}) = 0$$

Further simplification yields the ordinary differential equation

$$R'' + \left(\frac{2}{r} + \frac{F'_{b_0}}{F_{b_0}} \right) R' + \left(\frac{\sigma^2}{F_{b_0}^2} - \frac{l(l+1)}{r^2 F_{b_0}} \right) R = 0 \quad (1.2.2)$$

Near a horizon $r = r_h$, let $x = r - r_h$. Since

$$F_{b_0}(r) = F'_{b_0}(r_h)x + O(x^2),$$

and $F'_{b_0}(r_h) \neq 0$, the point $x = 0$ is a regular singular point of the radial equation. Let

$$c = F'_{b_0}(r_h).$$

The leading singular part of equation 1.2.2 is then

$$R'' + \frac{1}{x} R' + \frac{\sigma^2}{c^2 x^2} R = 0.$$

By the Frobenius theorem, the leading exponents of local solutions are determined by the indicial equation. Writing the leading term as

$$R(x) \sim x^\alpha,$$

we obtain

$$\alpha(\alpha-1)x^{\alpha-2} + \alpha x^{\alpha-2} + \frac{\sigma^2}{c^2} x^{\alpha-2} = 0.$$

Thus

$$\alpha^2 + \frac{\sigma^2}{c^2} = 0,$$

and hence the indicial roots are

$$\alpha_\pm = \pm \frac{i\sigma}{c}.$$

Therefore the two possible leading behaviours near the horizon are

$$R_\pm(r) \sim (r - r_h)^{\pm i\sigma/c}. \quad (1.2.3)$$

More precisely, the corresponding local Frobenius solutions have the form

$$R_\pm(r) = (r - r_h)^{\pm i\sigma/c} a_\pm(r, \sigma),$$

where $a_\pm$ are smooth and nonzero at $r = r_h$, after possibly restricting to a sufficiently small neighbourhood of the horizon.

These asymptotic behaviours already indicate why Sobolev regularity near the horizon is relevant. Indeed, for a local defining function $x \geq 0$, one has

$$x^\alpha \in H_{\text{loc}}^s \iff s < \text{Re}(\alpha) + \frac{1}{2}.$$

Thus the Sobolev membership of the two branches depends on $\text{Im}(\sigma)$. This will be made precise in Section 1.3, after introducing Regge–Wheeler coordinates, where the relation with the Fredholm threshold

$$\text{Im}(\sigma) > \frac{1 - 2s}{2\gamma}$$

will be explained.

1.3. The Regge Wheeler Coordinates and Optimality of Condition 1.1.3

By doing the Fourier transform with respect to t_* of the wave equation 1.1.1, we get

$$\hat{\square}_g(\sigma) = -F\partial_r^2 - \left(F' + \frac{2F}{r} \pm 2i\sigma\right) \partial_r \pm \frac{2i\sigma}{r} - r^{-2}\Delta_{\mathbb{S}^2}$$

Thus

$$P(\sigma) = -F^2\partial_r^2 - F\left(F' + \frac{2F}{r} \pm 2i\sigma\right) \partial_r \pm \frac{2iF\sigma}{r} - \frac{F}{r^2}\Delta_{\mathbb{S}^2}$$

We solve the homogeneous equation $P(\sigma)\hat{u} = 0$. Like in the earlier section, we do spherical decomposition to reduce it to the ODE

$$FR'' + \left(F' + \frac{2F}{r} \pm 2i\sigma\right) R' + \left(\pm \frac{2i\sigma}{r} - \frac{l(l+1)}{r^2}\right) R = 0. \quad (1.3.1)$$

Then using the Frobenius method as earlier, we get

$$\alpha(F'(r_h)\alpha \pm 2i\sigma) = 0$$

as the indicial equation which gives us the roots

$$\alpha_0 = 0 \quad \text{and} \quad \alpha_1 = \pm \frac{2i\sigma}{F'(r_h)}$$

where the sign of α_1 corresponds to the horizons.

So our first solution $R_0(r)$ is smooth near $r = r_h$ and the second solution has the leading behaviour $R_1(r) \sim (r - r_h)^{\pm \frac{2i\sigma}{F'(r_h)}}$.

Proposition 1.3.1. *Fix $\sigma \in \mathbb{C}$, a horizon $r_h \in \{r_-, r_+\}$, and an angular mode (ℓ, m) . Let $R_{\ell m}$ be a local radial solution of the separated equation 1.3.1 near $r = r_h$. Then the indicial equation has roots*

$$\alpha_0 = 0, \quad \alpha_1 = \pm \frac{2i\sigma}{F'(r_h)}.$$

The root $\alpha_0 = 0$ gives a smooth branch. The root α_1 gives the non-smooth indicial branch. Equivalently, after writing the exponent in terms of the surface gravity κ_h , the local solution can be written in the form

$$R_{\ell m}(r) = A_{h,\ell m}(\sigma)R_{h,\ell m,0}(r, \sigma) + B_{h,\ell m}(\sigma)(r - r_h)^{\frac{i\sigma}{\kappa_h}}R_{h,\ell m,1}(r, \sigma),$$

where $R_{h,\ell m,0}$ and $R_{h,\ell m,1}$ are smooth and nonzero at $r = r_h$. The second term is the non-smooth indicial branch. It belongs to H_{loc}^s near $r = r_h$ if and only if

$$\text{Im}(\sigma) < \kappa_h \left(\frac{1 - 2s}{2} \right).$$

This proposition and the subsequent statements are expanding on Haffner's statement in [10][Remark 5.8] and we will link them to the threshold condition given in 1.1.3.

Proof:

For notational simplicity, we suppress the dependence on (ℓ, m) and write $A_h, B_h, R_{h,0}$ and $R_{h,1}$. The Frobenius computation gives the indicial roots

$$\alpha_0 = 0, \quad \alpha_1 = \pm \frac{2i\sigma}{F'(r_h)}.$$

The root

$$\alpha_0 = 0$$

gives the smooth branch. Writing the second exponent in terms of the surface gravity κ_h gives the non-smooth branch

$$(r - r_h)^{\frac{i\sigma}{\kappa_h}} R_{h,1}(r, \sigma).$$

Since $R_{h,1}$ is smooth and nonzero at $r = r_h$, the Sobolev regularity of this branch is the same as that of

$$(r - r_h)^{i\sigma/\kappa_h}.$$

Let

$$\sigma = a + ib.$$

Then

$$\operatorname{Re}\left(\frac{i\sigma}{\kappa_h}\right) = -\frac{\operatorname{Im}(\sigma)}{\kappa_h}.$$

Using the standard criterion

$$x^\alpha \in H_{\text{loc}}^s \iff s < \operatorname{Re}(\alpha) + \frac{1}{2},$$

we obtain

$$(r - r_h)^{\frac{i\sigma}{\kappa_h}} \in H_{\text{loc}}^s \iff s < \frac{1}{2} - \frac{\operatorname{Im}(\sigma)}{\kappa_h}.$$

Equivalently,

$$\operatorname{Im}(\sigma) < \kappa_h \left(\frac{1 - 2s}{2} \right).$$

This proves the proposition.

Corollary 1.3.2. *Let*

$$\gamma = \max \left\{ \frac{1}{\kappa_-}, \frac{1}{\kappa_+} \right\}.$$

Fix $\sigma \in \mathbb{C}$ and an angular mode (ℓ, m) . If

$$\operatorname{Im}(\sigma) \geq \frac{1 - 2s}{2\gamma},$$

then H^s regularity excludes the non-smooth indicial branch at both horizons. Equivalently, in the local expansions at $r = r_-$ and $r = r_+$, one must have

$$B_{-\ell m}(\sigma) = B_{+\ell m}(\sigma) = 0.$$

Proof:

By Proposition 1.3.1, the non-smooth branch at $r = r_h$ belongs to H_{loc}^s precisely when

$$\operatorname{Im}(\sigma) < \kappa_h \left(\frac{1 - 2s}{2} \right).$$

Thus this branch is excluded at

$$r = r_h$$

when

$$\operatorname{Im}(\sigma) \geq \kappa_h \left(\frac{1-2s}{2} \right).$$

To exclude the non-smooth branch at both horizons, we require this condition for $h = -$ and $h = +$. Since $s \geq 1/2$, the factor $\frac{(1-2s)}{2}$ is non-positive. Therefore the condition is

$$\operatorname{Im}(\sigma) \geq \min(\kappa_-, \kappa_+) \left(\frac{1-2s}{2} \right).$$

But

$$\gamma = \max \left\{ \frac{1}{\kappa_-}, \frac{1}{\kappa_+} \right\} = \frac{1}{\min(\kappa_-, \kappa_+)},$$

so this becomes

$$\operatorname{Im}(\sigma) \geq \frac{1-2s}{2\gamma}.$$

This is exactly the threshold appearing in Theorem 1.1.1 with the exclusion of the equality. The radial point estimates later explain why the threshold 1.1.3 is precisely the condition excluding the bad horizon branch in both the horizons.

Remark 1.3.3. *The previous proposition and corollary were stated for a fixed angular mode (ℓ, m) . We now explain how the same conclusion applies to a full spherical harmonic expansion. Let I be a radial neighbourhood of a horizon and write*

$$u(r, \omega) = \sum_{\ell=0}^{\infty} \sum_{m=-\ell}^{\ell} u_{\ell m}(r) Y_{\ell m}(\omega),$$

where

$$u_{\ell m}(r) = \int_{\mathbb{S}^2} u(r, \omega) \overline{Y_{\ell m}(\omega)} d\omega.$$

We denote the map extracting the radial coefficient by

$$\pi_{\ell m} u = u_{\ell m}.$$

Since the spherical harmonics form an orthonormal basis of $L^2(\mathbb{S}^2)$, Parseval's identity gives

$$\|u\|_{L^2(I \times \mathbb{S}^2)}^2 = \sum_{\ell, m} \|u_{\ell m}\|_{L^2(I)}^2.$$

Thus $u \in L^2(I \times \mathbb{S}^2)$ if and only if each coefficient $u_{\ell m}$ belongs to $L^2(I)$ and

$$\sum_{\ell, m} \|\pi_{\ell m} u\|_{L^2(I)}^2 < \infty.$$

We next consider radial Sobolev regularity. Let $\chi \in C_0^\infty(I)$ be supported near the horizon. Using the definition of the restricted Sobolev space, extend χu to a function on $\mathbb{R}_r \times \mathbb{S}^2$. Define

$$\langle D_r \rangle^s = (1 + D_r^2)^{s/2}, \quad D_r = -i\partial_r.$$

If $u \in H^s(I \times \mathbb{S}^2)$, then

$$\langle D_r \rangle^s(\chi u) \in L^2(\mathbb{R} \times \mathbb{S}^2).$$

The projection $\pi_{\ell m}$ commutes with $\langle D_r \rangle^s$. Indeed, taking the Fourier transform in r , we obtain

$$\begin{aligned}\mathcal{F}_r(\pi_{\ell m}\langle D_r \rangle^s(\chi u))(\xi) &= \int_{\mathbb{S}^2} \langle \xi \rangle^s \widehat{\chi u}(\xi, \omega) \overline{Y_{\ell m}(\omega)} d\omega \\ &= \langle \xi \rangle^s \int_{\mathbb{S}^2} \widehat{\chi u}(\xi, \omega) \overline{Y_{\ell m}(\omega)} d\omega \\ &= \mathcal{F}_r(\langle D_r \rangle^s \pi_{\ell m}(\chi u))(\xi).\end{aligned}$$

Therefore

$$\pi_{\ell m}\langle D_r \rangle^s(\chi u) = \langle D_r \rangle^s \pi_{\ell m}(\chi u).$$

Applying Parseval's identity to $\langle D_r \rangle^s(\chi u)$ gives

$$\|\langle D_r \rangle^s(\chi u)\|_{L^2(\mathbb{R} \times \mathbb{S}^2)}^2 = \sum_{\ell, m} \|\langle D_r \rangle^s(\chi u_{\ell m})\|_{L^2(\mathbb{R})}^2.$$

Since the left-hand side is finite, we conclude that

$$\sum_{\ell, m} \|\langle D_r \rangle^s(\chi u_{\ell m})\|_{L^2(\mathbb{R})}^2 < \infty.$$

In particular,

$$u_{\ell m} \in H_{\text{loc}}^s(I)$$

for every fixed angular mode (ℓ, m) .

For each fixed mode, the Frobenius expansion near $r = r_h$ has the form

$$u_{\ell m}(r) = A_{h, \ell m}(\sigma) R_{h, \ell m, 0}(r, \sigma) + B_{h, \ell m}(\sigma) x_h^{\frac{i\sigma}{\kappa_h}} R_{h, \ell m, 1}(r, \sigma),$$

where x_h is a defining function of the horizon and $R_{h, \ell m, 0}$ and $R_{h, \ell m, 1}$ are smooth and nonzero at $r = r_h$.

Suppose that

$$x_h^{\frac{i\sigma}{\kappa_h}} R_{h, \ell m, 1}(r, \sigma) \notin H_{\text{loc}}^s.$$

Since $u \in H^s$, the preceding argument implies that $u_{\ell m} \in H_{\text{loc}}^s$. The smooth branch $R_{h, \ell m, 0}$ also belongs to H_{loc}^s . If $B_{h, \ell m}(\sigma) \neq 0$, then

$$x_h^{\frac{i\sigma}{\kappa_h}} R_{h, \ell m, 1}(r, \sigma) = \frac{1}{B_{h, \ell m}(\sigma)} (u_{\ell m} - A_{h, \ell m}(\sigma) R_{h, \ell m, 0})$$

would belong to H_{loc}^s , which is a contradiction. Hence

$$B_{h, \ell m}(\sigma) = 0.$$

Since this argument applies to every angular mode, we obtain

$$B_{h, \ell m}(\sigma) = 0 \quad \text{for all } (\ell, m).$$

If the non-smooth branch is allowed by the radial H^s condition, then a full spherical harmonic expansion must still satisfy the square-summability condition

$$\sum_{\ell, m} \left\| \langle D_r \rangle^s \left[\chi \left(A_{h, \ell m} R_{h, \ell m, 0} + B_{h, \ell m} x_h^{\frac{i\sigma}{\kappa_h}} R_{h, \ell m, 1} \right) \right] \right\|_{L^2(\mathbb{R})}^2 < \infty.$$

This is the summability condition obtained directly from the H^s regularity of the full solution. Thus the full spherical harmonic decomposition does not change the horizon threshold. When the non-smooth branch is excluded by H^s regularity, its coefficient vanishes for every angular mode. When it is allowed, the collection of radial modes must satisfy the square-summability condition above. Thus, if the full solution belongs to H^s , then every spherical harmonic coefficient has radial H_{loc}^s regularity and the corresponding radial Sobolev norms are square-summable. Proposition 1.3.1 may therefore be applied to each angular mode.

Proposition 1.3.4 (Optimality of Condition 1.1.3). *Let $s \geq 1/2$ and set*

$$\tau_s = \frac{1 - 2s}{2\gamma}.$$

Suppose that Condition 1.1.3 is replaced by

$$\text{Im}(\sigma) < \tau_s.$$

Then,

$$\hat{\square}_g(\sigma) : \mathcal{X}^s \rightarrow \mathcal{Y}^{s-1}$$

is not Fredholm. More precisely, its kernel is infinite dimensional.

Proof:

Let

$$c_s = \frac{1 - 2s}{2}.$$

Since $s \geq 1/2$, we have $c_s \leq 0$. Moreover,

$$\gamma = \max \left\{ \frac{1}{\kappa_-}, \frac{1}{\kappa_+} \right\} = \frac{1}{\min(\kappa_-, \kappa_+)},$$

and hence

$$\tau_s = \frac{1 - 2s}{2\gamma} = c_s \min(\kappa_-, \kappa_+).$$

Since $c_s \leq 0$,

$$c_s \min(\kappa_-, \kappa_+) = \max\{\kappa_- c_s, \kappa_+ c_s\}.$$

Therefore

$$\tau_s = \max \left\{ \kappa_- \left(\frac{1 - 2s}{2} \right), \kappa_+ \left(\frac{1 - 2s}{2} \right) \right\}.$$

Now choose σ such that

$$\text{Im}(\sigma) < \tau_s.$$

Since τ_s is the maximum of the two horizon thresholds, the inequality above implies that there is at least one horizon $r = r_h$ for which

$$\text{Im}(\sigma) < \kappa_h \left(\frac{1 - 2s}{2} \right).$$

By Proposition 1.3.1, the non-smooth indicial branch at this horizon belongs to H_{loc}^s . Hence H^s -regularity does not exclude this branch at $r = r_h$.

Now fix an angular mode (ℓ, m) . The separated radial equation is a second-order linear ordinary differential equation. Its space of solutions is two-dimensional.

At the horizon $r = r_h$, both the smooth branch and the non smooth branch are allowed by the H^s condition.

At the other horizon, H^s regularity imposes at most one linear condition. If the non-smooth branch there is not in H_{loc}^s , then its coefficient must vanish and if it is in H_{loc}^s , then no condition is imposed.

Thus, at the level of the separated radial equation, for each angular mode (ℓ, m) , there remains at least a one-dimensional space of radial solutions satisfying the H^s conditions at the two horizons. Hence, for each (ℓ, m) , we obtain a non-zero separated solution

$$u_{\ell m}(r, \omega) = R_{\ell m}(r) Y_{\ell m}(\omega)$$

such that

$$\widehat{\square}_g(\sigma)u_{\ell m} = 0.$$

Since the spherical harmonics $Y_{\ell m}$ are linearly independent, the corresponding solutions $u_{\ell m}$ are linearly independent. Therefore

$$\dim \ker \widehat{\square}_g(\sigma) = \infty.$$

But a Fredholm operator has finite-dimensional kernel. Hence the weakened condition cannot give a Fredholm operator for the full wave operator on the same H^s domain.

This proves that Condition 1.1.3 cannot be weakened in the above sense.

Remark 1.3.5. *The preceding argument uses the full spherical harmonic decomposition. This is essential since the infinite-dimensional kernel comes from allowing all angular modes (ℓ, m) . If one restricts the operator to a fixed eigenspace of $\Delta_{\mathbb{S}^2}$, then the above argument does not apply, since only finitely many angular modes are present.*

In the fixed angular momentum problem, the Fredholm property is not the obstruction in the same way. Instead, one expects a finite-dimensional kernel only at exceptional values of σ , so the issue becomes the failure of invertibility rather than the failure of Fredholmness. A precise treatment of this refined statement requires using the analyticity of the family $\widehat{\square}_g(\sigma)$ in σ , and we do not pursue it here.

Remark 1.3.6. *The previous argument explains why the Sobolev regularity has to be strong enough to exclude the non-smooth indicial branch. One may then ask why we do not impose an even stronger condition at the horizons. The reason is that the Sobolev condition excludes the non-smooth branch while keeping the smooth branch. No Sobolev space H^s is too strict in this sense, since a smooth function belongs to H^s for every s . To exclude also the smooth branch, one would have to impose a different type of condition, for example a weighted L^2 condition requiring decay at the horizons.*

Let

$$x_- = r - r_-, \quad x_+ = r_+ - r$$

be defining functions for the two horizons in the exterior region. Thus $x_- = 0$ at the black-hole horizon and $x_+ = 0$ at the cosmological horizon, while both are positive in the interior region $r_- < r < r_+$. For $\delta > 0$, consider the weighted space

$$L_\delta^2 = \{u : x_-^{-\delta} x_+^{-\delta} u \in L^2((r_-, r_+) \times \mathbb{S}^2)\}.$$

Let us examine precisely what this condition does to the indicial branches. Near one horizon, write $x = x_h$ for a local defining function. A branch of the form x^α , with $\alpha \in \mathbb{C}$, belongs to L_δ^2 near $x = 0$ if and only if

$$x^{-\delta} x^\alpha \in L^2(0, \epsilon).$$

Since

$$|x^\alpha| = x^{\operatorname{Re}(\alpha)}$$

for $x > 0$, this condition is equivalent to

$$\int_0^\epsilon x^{2\operatorname{Re}(\alpha) - 2\delta} dx < \infty.$$

Using the elementary criterion

$$\int_0^\epsilon x^\beta dx < \infty \iff \beta > -1,$$

we obtain

$$x^\alpha \in L_\delta^2 \iff 2\operatorname{Re}(\alpha) - 2\delta > -1.$$

Equivalently,

$$x^\alpha \in L_\delta^2 \iff \operatorname{Re}(\alpha) > \delta - \frac{1}{2}.$$

The smooth indicial branch corresponds to $\alpha = 0$. Indeed, this branch is smooth and nonzero at the horizon, so its leading term is a nonzero multiple of x^0 . Hence the smooth branch belongs to L_δ^2 precisely when

$$0 > \delta - \frac{1}{2},$$

or equivalently when $\delta < 1/2$. Therefore, if $\delta \geq 1/2$, the weighted space L_δ^2 already excludes the smooth branch.

The non-smooth branch near $r = r_h$ has the form

$$x_h^{\frac{i\sigma}{\kappa_h}} R_{h,1}(r, \sigma),$$

where $R_{h,1}$ is smooth and nonzero at $r = r_h$. Since multiplication by such a smooth nonzero factor does not affect the local weighted L^2 condition, the relevant exponent is

$$\alpha_h = \frac{i\sigma}{\kappa_h}.$$

Writing $\sigma = a + ib$, we have

$$\operatorname{Re}\left(\frac{i\sigma}{\kappa_h}\right) = -\frac{\operatorname{Im}(\sigma)}{\kappa_h}.$$

Thus the non smooth branch belongs to L_δ^2 precisely when

$$-\frac{\operatorname{Im}(\sigma)}{\kappa_h} > \delta - \frac{1}{2}.$$

Equivalently, it is excluded from L_δ^2 whenever

$$\delta \geq \frac{1}{2} - \frac{\operatorname{Im}(\sigma)}{\kappa_h}.$$

Therefore, by choosing δ sufficiently large, we can exclude both the smooth and the non-smooth indicial branches at both horizons.

Let us now see why such a condition is too strong. Fix an angular mode (ℓ, m) . The separated radial equation is a second-order linear ODE. Near each horizon, every local solution is a linear combination of the two indicial branches:

$$R(r) = A_h(\sigma) R_{h,0}(r, \sigma) + B_h(\sigma) x_h^{\frac{i\sigma}{\kappa_h}} R_{h,1}(r, \sigma).$$

If the space L_δ^2 excludes both branches at a horizon, then a solution belonging to L_δ^2 must have

$$A_h(\sigma) = 0, \quad B_h(\sigma) = 0.$$

Thus the solution vanishes near that horizon. By uniqueness for the radial ODE, it must then vanish identically. Hence this overly strong weighted condition kills the homogeneous solutions and in particular, it gives a trivial kernel for the separated problem.

However, this does not give a good Fredholm problem, because the cokernel becomes too large. To see this, consider the operator

$$P_\sigma = \widehat{\square}_g(\sigma)$$

acting on a weighted domain based on L_δ^2 . The dual of L_δ^2 with respect to the unweighted L^2 pairing is $L_{-\delta}^2$, where

$$L_{-\delta}^2 = \{v : x_-^\delta x_+^\delta v \in L^2((r_-, r_+) \times \mathbb{S}^2)\}.$$

This dual space is much larger near the horizons. Indeed, near a horizon, a function behaving like x^β belongs to $L^2_{-\delta}$ when

$$\int_0^\epsilon x^{2\operatorname{Re}(\beta)+2\delta} dx < \infty,$$

which is equivalent to

$$\operatorname{Re}(\beta) > -\delta - \frac{1}{2}.$$

For large δ , this condition allows many singular behaviours. In particular, if δ has been chosen large enough, the adjoint indicial branches are allowed in the dual space $L^2_{-\delta}$.

Consequently, for each angular mode (ℓ, m) , one obtains a non-zero separated solution of the adjoint homogeneous equation

$$P_\sigma^* v_{\ell m} = 0, \quad v_{\ell m}(r, \omega) = S_{\ell m}(r) Y_{\ell m}(\omega),$$

which belongs to the dual weighted space $L^2_{-\delta}$. Now suppose that f lies in the range of P_σ acting on the too restrictive weighted domain. Then

$$f = P_\sigma u$$

for some u satisfying the strong weighted condition. Pairing with $v_{\ell m}$ gives

$$\langle f, v_{\ell m} \rangle = \langle P_\sigma u, v_{\ell m} \rangle.$$

After integration by parts, the boundary terms vanish because u decays strongly at the horizons. Since $P_\sigma^* v_{\ell m} = 0$, we obtain

$$\langle f, v_{\ell m} \rangle = \langle u, P_\sigma^* v_{\ell m} \rangle = 0.$$

Thus each adjoint solution $v_{\ell m}$ gives a compatibility condition on the range:

$$\langle f, v_{\ell m} \rangle = 0.$$

Since there are infinitely many angular modes and the spherical harmonics are linearly independent, these compatibility conditions are independent for different (ℓ, m) . Therefore the cokernel is infinite-dimensional.

This shows that imposing a condition which is too strong may indeed kill the kernel, but it destroys Fredholmness by producing an infinite-dimensional cokernel. The Sobolev threshold in Condition 1.1.3 is therefore balanced as it excludes the non smooth horizon branch while retaining the smooth horizon crossing branch.

Remark 1.3.7. The coordinate t_* is not uniquely defined. Häfner only requires it to agree with $t + r_*$ near the blackhole horizon and $t - r_*$ near the cosmological horizon with a smooth interpolation in between. We now check that the Sobolev threshold found in Proposition 1.3.1 does not depend on this arbitrary interpolation. Thus the threshold is geometric: it depends only on the surface gravities of the two horizons.

Suppose t_* and $\tilde{t}_*$ are as follows:

$$\tilde{t}_* = t_* + a(r)$$

where $a(r)$ is smooth.

Now consider

$$u = e^{-i\sigma t_*} R(r, \omega).$$

Then using the new coordinates, we get

$$\begin{aligned} u &= e^{-i\sigma(\tilde{t}_* - a(r))} R(r, \omega) \\ &= e^{-i\sigma \tilde{t}_*} e^{i\sigma a(r)} R(r, \omega) \end{aligned}$$

which gives us

$$R(r, \omega) = e^{i\sigma a(r)} R(r, \omega)$$

Since $a(r)$ is smooth, $e^{i\sigma a(r)}$ is smooth and never zero and thus preserves Sobolev regularity when multiplied. More precisely,

R belongs to H^s locally if and only if $e^{i\sigma a(r)} R$ belongs to H^s .

This is true because multiplication by a smooth function is bounded on H^s , and multiplication by its inverse is also bounded on H^s . Therefore the Sobolev membership of the bad branch does not change. The branch may be multiplied by a smooth factor, but its singular power at the horizon is the same.

We can thus conclude that the threshold $s = \frac{1}{2} - \frac{\text{Im}(\sigma)}{\kappa_h}$ does not depend on the chosen interpolation in t_* . It depends only on the surface gravity κ_h . Thus the threshold is geometric and not coordinate based.

2. Proof of Theorem 1.1.1

The goal of this section will be to provide the tools to prove Theorem 1.1.1. We will begin with the recollection of a few preliminary results of microlocal analysis and then develop on the ideas of the proof through subsequent sections before giving a detailed proof of the theorem.

2.1. Microlocal Analysis

We will follow Häfner's [10] notations and presentation for this section. Let us start with the differential operator of order m on $\mathbb{R}^n$.

$$A = \sum_{|\alpha| \leq m} a_\alpha(x) D_x^\alpha \in \text{Diff}^m(\mathbb{R}^\times)$$

Then using the inverse Fourier transform, we can write it as

$$(Au)(x) = \sum_{|\alpha| \leq m} a_\alpha(x) \mathcal{F}^{-1}(\xi^\alpha (\mathcal{F}u)(\xi)) = \frac{1}{(2\pi)^n} \int e^{i\langle x, \xi \rangle} \sum_{|\alpha| \leq m} a_\alpha(x) \xi^\alpha \widehat{u}(\xi) d\xi := (Op(a)u)(x)$$

where $a(x, \xi) = \sum_{|\alpha| \leq m} a_\alpha(x) \xi^\alpha$ is the symbol of the operator.

A smooth function a on $T^*\mathbb{R}^n$ is a symbol of order m if

$$|\partial_x^\alpha \partial_\xi^\beta a(x, \xi)| \leq C_{\alpha, \beta} \langle \xi \rangle^{m-\beta}.$$

For a general symbol $a(x, \xi)$, we can define the corresponding pseudodifferential operator as

$$(Op(a)u)(x) = \frac{1}{(2\pi)^n} \int e^{ix\xi} a(x, \xi) \widehat{u}(\xi) d\xi = \frac{1}{(2\pi)^n} \int \int e^{i\langle x-y, \xi \rangle} a(x, \xi) u(y) dy d\xi \quad (2.1.1)$$

We denote by $S^m(\mathbb{R}^n)$ the set of symbols of order m and for any $a \in S^m$, we define $Op(a)$ by 2.1.1. We denote by

$$\Psi^m(\mathbb{R}^n) = \{Op(a) : a \in S^m(\mathbb{R}^n)\}$$

the set of pseudodifferential operators of order m . Further, for any pseudodifferential operator of order m , we define its principal symbol as

$$\sigma_m(A) := a(x, \xi) \bmod S^{m-1}$$

and this is independent of choice of coordinates and any quantization we use.

Theorem 2.1.1. [10] *Theorem 4.1[Fundamental properties of pseudodifferential operators]*

1. *Algebra property:* Ψ^m is a vector space for each $m \in \mathbb{R}$. If $A \in \Psi^m$ and $B \in \Psi^{m'}$, then $AB \in \Psi^{m+m'}$ and $A^* \in \Psi^m$. The composition of operators is associative and distributive. The identity operator is in $\Psi^0(X)$.
2. *Boundedness:* If $A \in \Psi^m$, then $A \in \mathcal{L}(H^s, H^{s-m})$ for all $s \in \mathbb{R}$.
3. *Principal symbol homomorphism:*

$$\sigma_{m+m'}(AB) = \sigma_m(A)\sigma_{m'}(B), \quad \sigma_m(A^*) = \overline{\sigma_m(A)}.$$

4. *Symbol of commutator:* If $A \in \Psi^m(X)$ and $B \in \Psi^{m'}(X)$, then $[A, B] \in \Psi^{m+m'-1}(X)$ and

$$\sigma_{m+m'-1}([A, B]) = i\{\sigma_m(A), \sigma_{m'}(B)\} = H_a b \quad \text{where } a = \sigma_m(A), \ b = \sigma_{m'}(B).$$

Here $\{\cdot, \cdot\}$ is the Poisson bracket and H_a is the Hamiltonian vector field associated to a .

Theorem 2.1.2 ([10] Theorem 4.3). *Let $P \in \Psi^m$ be a uniformly elliptic operator. Then there exists $Q \in \Psi^{-m}$ such that*

$$QP - 1 \in \Psi^{-\infty} \quad , \quad PQ - 1 \in \Psi^{-\infty}$$

Let us now state an important inequality which is relied upon extensively through the proof of Theorem 1.1.1.

Theorem 2.1.3 ([10] Theorem 4.4). *[Sharp Gårding inequality] Let $a \in S^{2m+1}$ and $\text{Re}(a) \geq 0$. Then for $u \in \mathcal{S}$, we have*

$$\text{Re}\langle \text{Op}(a)u, u \rangle \geq -C\|u\|_{H^m}^2.$$

We now want to define pseudodifferential calculus on a compact manifold such that the above properties remain valid.

We begin by first radially compactifying each fibre of $T^*\mathbb{R}^n$ to obtain $\overline{T^*\mathbb{R}^n}$ and denote by $S\mathbb{R}^n$ the cosphere bundle $\partial\overline{T^*\mathbb{R}^n}$. The idea is to map $\mathbb{R}_\xi^n$ to the closed unit ball $\bar{B}_n$. The reason we do this is because this way, points near the origin stay near the origin whereas as $|\xi| \rightarrow \infty$, points are mapped to the boundary of the ball.

A classical symbol of order m is $\langle \xi^m \rangle$ times a smooth function on $\overline{T^*\mathbb{R}^n}$ and for a classical symbol a , we can write down Taylor's series expansion near $\frac{1}{|\xi|} = 0$ to obtain

$$a(x, \xi) \sim \sum_{j=0}^{\infty} a_{m-j}(x, \hat{\xi}) |\xi|^{m-j}, \quad a_{m-j} \in C^\infty(\mathbb{R}^n \times \mathbb{S}^{n-1}) \quad \text{and} \quad \hat{\xi} = \frac{\xi}{|\xi|} \in \mathbb{S}^{n-1}$$

Now suppose $A \in \Psi^m$ has compactly supported Schwartz Kernel supported in $U \times U$. Then under a change of coordinates, $\kappa : U \mapsto V$, the principal symbol of an operator transforms as a function on the cotangent bundle and the Hamiltonian field is invariantly defined under this change of coordinates.

This allows us to define pseudodifferential calculus on a compact manifold X without the loss of above properties. In the non compact case, we have to make sure that the operators are properly supported or impose conditions at infinity.

2.2. The Wave Equation on Schwarzschild De Sitter metric

Recall that $M = \mathbb{R}_{t_*} \times X$ where $X = (r_- - \epsilon, r_+ + \epsilon) \times S_\omega^2$. Using the notation $D_x = -i\partial_x$ we can write the wave operator as

$$\square_g = \begin{cases} 2Dt_*D_r + \frac{1}{r^2}D_r r^2 F D_r - \frac{1}{r^2}\Delta_{S^2} & \text{near } r = r_-, \\ -2Dt_*D_r + \frac{1}{r^2}D_r r^2 F D_r - \frac{1}{r^2}\Delta_{S^2} & \text{near } r = r_+. \end{cases}$$

and after doing a Fourier transform with respect to t_* , i.e. using

$$\widehat{\square}_g(\sigma) = e^{it_*\sigma} \square_g e^{-it_*\sigma}$$

we get

$$\widehat{\square}_g(\sigma) = \begin{cases} -2\sigma D_r + \frac{1}{r^2}D_r r^2 F D_r - \frac{1}{r^2}\Delta_{S^2} & \text{for } r = r_-, \\ 2\sigma D_r + \frac{1}{r^2}D_r r^2 F D_r - \frac{1}{r^2}\Delta_{S^2} & \text{for } r = r_+. \end{cases}$$

We will now consider the problem $\square_g u = f$ where f is a smooth and compactly supported function on M and with the requirement that $\text{supp } u \subset \{t_* \geq -T\}$ for some $T > 0$.

2.3. Tools and Principle Ideas of the Proof

Lemma 2.3.1 ([10] Lemma 5.5). *Let $X_0 \subset X_1 \subset X_2$ and $Y_0 \subset Y_1 \subset Y_2$ be Banach Spaces with continuous dense inclusions. Let $P : X_1 \rightarrow Y_2$ be a bounded operator such $P|_{X_0}$ is bounded from X_0 to Y_1 . Assuming both inclusions $X_1 \subset X_2$ and $Y_0 \subset Y_1$ are compact and there exists $C > 0$ such that for every $u \in X_1$ and for all $v \in Y_1^*$, we have*

$$\|u\|_{X_1} \leq C (\|Pu\|_{Y_1} + \|u\|_{X_2}) \quad (2.3.1)$$

$$\|v\|_{Y_1^*} \leq C (\|P^*v\|_{X_1^*} + \|v\|_{Y_0^*}) \quad (2.3.2)$$

Then, P is Fredholm as an operator between the Banach space $\mathcal{A} = \{u \in X_1 : Pu \in Y_1\}$ and Y_1 where $\mathcal{A}$ is endowed with the norm $\|u\|_{\mathcal{A}}^2 = \|u\|_{X_1}^2 + \|Pu\|_{Y_1}^2$.

We now introduce the following notations:

Let $\tilde{X}$ be a smooth manifold without boundary, and let $X \subset \tilde{X}$ be an open domain whose closure $\overline{X}$ is a smooth manifold with boundary. We define the extendible Sobolev space on X by

$$\bar{H}^s(X) = \{u \in \mathcal{D}'(X^\circ) : \text{there exists } \tilde{u} \in H^s(\tilde{X}) \text{ such that } \tilde{u}|_{X^\circ} = u\}.$$

The norm is given by

$$\|u\|_{\bar{H}^s(X)} = \inf\{\|\tilde{u}\|_{H^s(\tilde{X})} : \tilde{u}|_{X^\circ} = u\}.$$

Thus $\bar{H}^s(X)$ consists of distributions on X° which extend as H^s distributions across the boundary of X . Similarly, we define the supported Sobolev space by

$$\dot{H}^s(X) = \{u \in H^s(\tilde{X}) : \text{supp } u \subset \overline{X}\}.$$

Thus $\dot{H}^s(X)$ consists of H^s -distributions on $\tilde{X}$ supported in $\overline{X}$.

Ideas of the Proof

We wish to use Lemma 2.3.1 to show that $\hat{\square}_g(\sigma) : \mathcal{X}^s \rightarrow \mathcal{Y}^{s-1}$ is Fredholm. In order to do that, we make estimates of the form

$$\|u\|_{\bar{H}^s(X)} \leq C \left(\|\hat{\square}_g(\sigma)u\|_{\bar{H}^{s-1}(X)} + \|u\|_{\bar{H}^{s_0}(X)} \right)$$

$$\|u\|_{\dot{H}^s(X)} \leq C \left(\|\hat{\square}_g^*(\sigma)u\|_{\dot{H}^{s-1}(X)} + \|u\|_{\dot{H}^{s_0}(X)} \right).$$

In order to make such estimates, we will have to look differently at the set of elliptic, radial and hyperbolic points, make estimates there and then glue them all together.

Using standard energy estimates, we know that $|u(t_*, r, \omega)| \lesssim e^{Ct_*}$. We claim that the Fourier transform of u is holomorphic on $\text{Im}(\sigma) > C$.

Proof:

Let $\sigma = \sigma_R + i\sigma_I$

$$\hat{u}(\sigma) = \int_{t_0}^{\infty} e^{i\sigma t_*} u(t_*) dt_*$$

Let $f(t_*, \sigma) = e^{i\sigma t_*} u(t_*)$. Then

$$|f(t_*, \sigma)| = e^{-\sigma_I t_*} |u(t_*)|$$

We know there exists $A > 0$ and $C > 0$ such that $|u(t_*)| \leq Ae^{Ct_*}$. Let $\text{Im}(\sigma) \geq C + \epsilon$ for some $\epsilon > 0$. Then

$$\sigma_I \geq C + \epsilon \quad \text{or} \quad -\sigma_I \leq -C - \epsilon$$

So,

$$\begin{aligned}
\int_{t_0}^{\infty} |e^{i\sigma t_*} u(t_*) dt_*| &= \int_{t_0}^{\infty} |e^{-i\sigma_I t_*} u(t_*) dt_*| \\
&\leq \int_{t_0}^{\infty} |e^{-(C+\epsilon)t_*} C A e^{C t_*} dt_*| \\
&= \frac{e^{-\epsilon t_0}}{\epsilon} \\
&< \infty
\end{aligned}$$

Since $f(t_*, \sigma)$ is continuous in σ for each t_* and the integral converges uniformly on compact subsets, $\hat{u}(\sigma)$ is a continuous function of σ on $\text{Im}(\sigma) > C$.

It remains to show complex differentiability. For each fixed t_* ,

$$\frac{\partial}{\partial \sigma} (e^{i\sigma t_*} u(t_*)) = i t_* e^{i\sigma t_*} u(t_*).$$

Let $K \subset \{\text{Im}(\sigma) > C\}$ be compact. Then there exists $\epsilon > 0$ such that

$$\text{Im}(\sigma) \geq C + \epsilon \quad \text{for all } \sigma \in K.$$

Hence

$$|i t_* e^{i\sigma t_*} u(t_*)| \leq A |t_*| e^{-\epsilon t_*}.$$

Since

$$\int_{t_0}^{\infty} A |t_*| e^{-\epsilon t_*} dt_* < \infty,$$

we may differentiate under the integral sign. Therefore

$$\hat{u}'(\sigma) = \int_{t_0}^{\infty} i t_* e^{i\sigma t_*} u(t_*) dt_*.$$

Thus $\hat{u}$ is holomorphic on $\text{Im}(\sigma) > C$.

Let us now consider the unitary transformation

$$U : L^2((r_-, r_+) \times \mathbb{S}^2, r^2 dr d\omega) \rightarrow L^2((r_-, r_+) \times \mathbb{S}^2, dr d\omega)$$

$$u \mapsto r u$$

Near $r = r_{\pm}$, we have

$$r \widehat{\square_g(\sigma)} r^{-1} = r^{-1} D_r r^2 F D_r r^{-1} \pm 2\sigma r^{-1} D_r r - r^{-2} \Delta_{\mathbb{S}^2}$$

which has the principal classical symbol $\sigma_2(\widehat{\square_g(\sigma)}) = F\xi^2 + r^{-2}|\eta|^2 := \rho$.

This operator is elliptic all over the region $r_- < r < r_+$ and from now, we will work with this operator.

2.4. Elliptic Region Estimates

Proposition 2.4.1 ([10] Proposition 5.6). *Let $\chi \in C_0^\infty(r_- + \delta, r_+ - \delta)$ for some $\delta > 0$. Then we have the estimate*

$$\|\chi u\|_{\bar{H}^s} \lesssim \|\widehat{\square_g(\sigma)} u\|_{\bar{H}^{s-2}} + \|u\|_{\bar{H}^{s-1}} \quad (2.4.1)$$

We will expand on the proof given in [10] and precisely state which tools were used for the proof.

Proof:

Let $\phi \in C_0^\infty((-1, 1))$ with $\phi = 1$ in a neighborhood of 0. Then

$$\chi(x) = \chi(x)\phi(\rho) + \chi(x)(1 - \phi(\rho))$$

where

$$\rho(x, \zeta) = \rho(r, \omega, \xi, \eta) = F\xi^2 + r^{-2}|\eta|^2.$$

We note that $\chi(x)\phi(\rho) \in S^{-\infty}$ and $\chi(x)(1 - \phi(\rho)) \in S^0$.

We first prove the estimate for $s = 0$.

Let $\phi \in C_0^\infty((-1, 1))$ be such that $\phi = 1$ near 0.

At the level of symbols, we have

$$\chi = \chi\phi(\rho) + \chi(1 - \phi(\rho)).$$

Therefore, by the linearity of the quantization map,

$$Op(\chi(x)) = Op(\chi(x)\phi(\rho)) + Op(\chi(x))(1 - \phi(\rho)) \quad (2.4.2)$$

The first term is residual. Indeed, on $\text{supp } \chi$, the symbol ρ is elliptic, and therefore the condition $\phi(\rho) \neq 0$ restricts (ξ, η) to a compact set. Thus

$$Op(\chi\phi(\rho)) \in \Psi^{-\infty}.$$

Consequently,

$$\|Op(\chi\phi(\rho))u\|_{\bar{H}^s} \lesssim \|u\|_{\bar{H}^{s-1}}.$$

Thus we can ignore the first term of equation 2.4.2. Further, through the use of rule of composition given in Theorem 2.1.1, we can get

$$Op(\chi(x)(1 - \phi(\rho))) = Op\left(\chi(x)\frac{1 - \phi(\rho)}{\rho}\right)\hat{\square}_g(\sigma) + R, \quad R \in \Psi^{-1} \quad (2.4.3)$$

Then for $s = 0$.

$$\begin{aligned} \|\chi(x)u\|_{L^2} &\leq \|Op(\chi(x)\phi(\rho))u\|_{L^2} + \|Op(\chi(x)(1 - \phi(\rho)))u\|_{L^2} \\ &\lesssim \|\hat{\square}_g(\sigma)u\|_{H^{-2}} + \|u\|_{H^{-1}} \end{aligned}$$

Now, $\Lambda^2 := Op(\xi^2 + |\eta|^2)$ belongs to Ψ^2 . Note that

$$\|v\|_{\bar{H}^s} = \|\Lambda^s v\|_{L^2}$$

From the $s = 0$ case, we know that

$$\begin{aligned} \|\chi\Lambda^s u\|_{L^2} &\lesssim \|\hat{\square}_g(\sigma)\Lambda^s u\|_{\bar{H}^{-2}} + \|\Lambda^s u\|_{\bar{H}^{-1}} \\ &= \|\hat{\square}_g(\sigma)\Lambda^s u\|_{\bar{H}^{-2}} + \|u\|_{\bar{H}^{s-1}} \end{aligned} \quad (2.4.4)$$

Also,

$$\hat{\square}_g(\sigma)\Lambda^s u = \Lambda^s \hat{\square}_g(\sigma)u + [\hat{\square}_g(\sigma), \Lambda^s]u$$

Thus for $s \neq 0$ we get that

$$\begin{aligned} \|\hat{\square}_g(\sigma)\Lambda^s u\|_{H^{-2}} &\leq \|\Lambda^s \hat{\square}_g(\sigma)u\|_{H^{-2}} + \|[\hat{\square}_g(\sigma), \Lambda^s]u\|_{H^{-2}} \\ &\lesssim \|\hat{\square}_g(\sigma)u\|_{\bar{H}^{s-2}} + \|u\|_{\bar{H}^{s-1}} \end{aligned} \quad (2.4.5)$$

where the final estimate follows from the fact that $[\hat{\square}_g(\sigma), \Lambda^s] \in \Psi^{2+s-1} = \Psi^{s+1}$ and an operator of order $s+1$ maps H^{s-1} to H^{-2} . Thus we can conclude the proof through equations 2.4.4 and 2.4.5.

2.5. Radial Point Analysis

Our goal in this section is to look at the following proposition:

Proposition 2.5.1 ([10] Proposition 5.7). *Suppose that $u \in \bar{H}^s$ with $s > \frac{1}{2} - \frac{1}{\kappa_{\pm}} \text{Im}(\sigma)$. Let $\chi_{\pm} \in C_0^\infty((r_{\pm} - \delta, r_{\pm} + \delta))$. Then for $\delta > 0$ sufficiently small, we have the estimate*

$$\|\chi_{\pm} u\|_{\bar{H}^s} \lesssim \|\widehat{\square}_g(\sigma)u\|_{\bar{H}^{s-1}} + \|u\|_{\bar{H}^{s-\frac{1}{2}}} \quad (2.5.1)$$

The details of the proof for this proposition can be found in Häfner [10] and the references cited there. We will not write it in this thesis but instead look at a similar model problem which will make the ideas of the proof given in Häfner [10] technically simpler. Before we describe our model problem, let us give a general recipe of how radial point estimates are made in our setting.

We will closely follow the works of Fang-Sussman [9] and Häfner [10].

Let $P \in \Psi^m$ be a pseudodifferential operator with symbol ρ . The characteristic set is

$$\Sigma = \{(x, \xi) \in T^* \setminus 0 : \rho(x, \xi) = 0\}$$

and the Hamiltonian vector field

$$H_\rho = \sum_j \partial_{\xi_j} \rho \partial_{x_j} - \partial_{x_j} \rho \partial_{\xi_j}$$

generates the bicharacteristic flow. Away from Σ , elliptic point estimates apply. At non-radial characteristic points, where $H_\rho \neq 0$, one uses propagation of singularities. Radial point analysis is needed at characteristic points where H_ρ becomes radial in the fibres or equivalently when

$$H_\rho = C \sum_j \xi_j \partial_{\xi_j}$$

along a conic subset of Σ . After radial compactification of T^*X such points become fixed points of the rescaled Hamilton flow. Fang-Sussman [9] describe radial point estimates as positive commutator estimates near these simple zeros of the compactified Hamilton vector field.

The basic tool is the positive commutator identity. If $A = A^*$ is a pseudodifferential microlocalizer, then

$$2i \text{Im} \langle Pu, Au \rangle = \langle [A, P]u, u \rangle + \langle (P - P^*)Au, u \rangle. \quad (2.5.2)$$

The principal symbol of the commutator is given by $H_\rho a$ where $a = \sigma(A)$ as given in Theorem 2.1.1. So we choose A such that $H_\rho a$ along with the contribution from $P - P^*$ has a definite sign. Fang-Sussman [9] emphasize that the flexibility of taking A to be a pseudodifferential operator is essential because it allows localization both in position and in frequency.

The general way of choosing a commutant for a pseudodifferential operator P of order m is to take the symbol

$$c = |\xi|^{s-\frac{m-1}{2}} \times \text{cut offs}$$

and then let $A = C^*C$ where $C = Op(c)$.

After the choice has been made, we look at the identity 2.5.2 and make estimates under certain conditions based on whether the operators are self adjoint or not and use tools such the Garding Inequality 2.1.3, Cauchy Schwarz estimates or other properties of pseudodifferential operators.

The distinction between sources and sinks is important. At a source, bicharacteristics flow away from the radial set. With a cutoff chosen monotone along the flow, the derivative terms from $H_\rho \chi$ have a favorable sign, and one obtains a source estimate of the form

$$\|Bu\|_{H^s} \lesssim \|Pu\|_{H^{s-m+1}} + \|u\|_{H^{s-\frac{m-1}{2}}}$$

At a radial sink, the direct estimate usually has a different form. Since bicharacteristics flow into the radial set, one generally needs an additional term measuring regularity in a punctured neighborhood through which the flow enters the sink:

$$\|Bu\|_{H^s} \lesssim \|Pu\|_{H^{s-m+1}} + \|Ru\|_{H^s} + \|u\|_{H^{s-\frac{m-1}{2}}}$$

where R is supported in a punctured neighborhood of the sink.

Model Problem

We work on

$$I = (-1, 1) \quad \text{and the operator} \quad P_\sigma = D_x x D_x - \sigma D_x \quad (2.5.3)$$

It is clear that the principal symbol is given by

$$p = x\xi^2 \quad (2.5.4)$$

and the characteristic set is

$$\Sigma = \{(x, \xi) \in T^*I \setminus \{0\} : p(x, \xi) = 0\}$$

Since $\xi \neq 0$ on $T^*I \setminus \{0\}$, we get

$$\Sigma = \{x = 0, \xi \neq 0\}$$

It has two connected pieces

$$\Sigma_+ = \{x = 0, \xi > 0\} \quad \text{and} \quad \Sigma_- = \{x = 0, \xi < 0\}.$$

Now,

$$\begin{aligned} H_p &= \partial_\xi p \partial_x - \partial_x p \partial_\xi \\ &= 2x\xi \partial_x - \xi^2 \partial_\xi \end{aligned} \quad (2.5.5)$$

On the characteristic set, since $x = 0$,

$$H_p = -\xi^2 \partial_\xi$$

and it is thus radial at every point of Σ . Thus both Σ_+ and Σ_- are radial sets. To see the source/sink structure, we compactify the fibrers.

On $\xi > 0$, we set

$$\rho = \xi^{-1}.$$

Then near $\rho = 0$, the rescaled Hamilton vector field is

$$\rho H_p = 2x \partial_x + \rho \partial_\rho.$$

Thus the flow of ρH_p is given by

$$x(t) = e^{2t} x(0), \quad \rho(t) = e^t \rho(0).$$

Hence

$$(x(t), \rho(t)) \longrightarrow (0, 0) \quad \text{as } t \rightarrow -\infty.$$

Therefore bicharacteristics starting near Σ_+ converge to Σ_+ in backward time. Equivalently, Σ_+ is a radial source for the forward Hamilton flow. This backward convergence is the relevant property in the source estimate.

Similarly, on $\xi < 0$, we set

$$\rho = -\xi^{-1} > 0.$$

Then

$$\rho H_p = -2x\partial_x - \rho\partial_\rho.$$

Thus

$$x(t) = e^{-2t}x(0), \quad \rho(t) = e^{-t}\rho(0),$$

and hence

$$(x(t), \rho(t)) \longrightarrow (0, 0) \quad \text{as } t \rightarrow +\infty.$$

Therefore Σ_- is a radial sink for the forward Hamilton flow.

Proposition 2.5.2. *Let*

$$s_0 = \frac{1}{2} - \text{Im}(\sigma).$$

Let $s > s_0$, and let $\chi, \tilde{\chi} \in C_0^\infty((-1, 1))$ be such that $\tilde{\chi} = 1$ on $\text{supp}\chi$ and $\text{supp}\chi$ is sufficiently close to 0. Assume that u has a priori microlocal regularity above the threshold near $\text{supp}\tilde{\chi}$, namely

$$\tilde{\chi}u \in H^{s'} \quad \text{for } s' \in [s - \frac{1}{2}, s].$$

Assume also that

$$\tilde{\chi}P_\sigma u \in H^{s-1}.$$

Then

$$\|\chi u\|_{H^s} \leq C \left(\|\tilde{\chi}P_\sigma u\|_{H^{s-1}} + \|\tilde{\chi}u\|_{H^{s-\frac{1}{2}}} \right). \quad (2.5.6)$$

An analogous claim followed by a brief proof will be made later in the sink case.

Proof of Proposition 2.5.2

Choose $\chi \in C_0^\infty((-1, 1))$ such that

$$\chi \geq 0, \quad \chi = 1 \quad \text{near } 0 \quad \text{and} \quad x\chi'(x) \leq 0.$$

So χ decreases as $|x|$ increases.

Choose $\psi \in C^\infty(\mathbb{R})$ such that

$$\psi(t) = \begin{cases} 1 & \text{if } t \geq 2 \\ 0 & \text{if } t \leq 1 \end{cases} \quad (2.5.7)$$

Define

$$c(x, \xi) = \chi(x)\psi(\xi)|\xi|^{s-\frac{1}{2}} \quad (2.5.8)$$

for $|\xi| \geq 1$ as a symbol in $S^{s-\frac{1}{2}}$.

Let

$$C = Op(c) \in \Psi^{s-\frac{1}{2}} \quad (2.5.9)$$

and define the commutant

$$A = C^*C \quad (2.5.10)$$

Thus A has principal symbol

$$a(x, \xi) = \chi(x)^2\psi(\xi)^2\xi^{2s-1}$$

The reason for the exponent $2s - 1$ is important. Since P_σ has order 2, the commutator

$$[P_\sigma, A]$$

has order

$$2 + (2s - 1) - 1 = 2s$$

which helps in the control of the H^s norm.

Since $D_x x D_x$ is formally self adjoint and $P_\sigma^* = D_x x D_x - \bar{\sigma} D_x$,

$$P_\sigma^* - P_\sigma = 2i \operatorname{Im}(\sigma) D_x$$

By commutator identity 2.5.2, we get

$$2\operatorname{Im}\langle P_\sigma u, Au \rangle = \langle (i[P_\sigma, A] + i(P_\sigma^* - P_\sigma)A)u, u \rangle \quad (2.5.11)$$

If we let

$$Q = i[P_\sigma, A] + i(P_\sigma^* - P_\sigma)A \quad (2.5.12)$$

then the principal symbol of Q is

$$q := \sigma_{2s}(Q) = H_p a - 2\operatorname{Im}(\sigma)\xi a \quad (2.5.13)$$

Now,

$$H_p a = -(2s-1)\chi^2\psi^2\xi^{2s} - 2\chi^2\psi\psi'\xi^{2s+1} + 4x\chi\chi'\psi^2\xi^{2s} \quad (2.5.14)$$

The second term is smoothing and can be ignored whereas the third term here is non positive. The principal symbol of $i[P_\sigma, A]$ is $H_p a$ and so using equations 2.5.11, 2.5.13 and 2.5.14, we get

$$q = (-(2s-1)\chi^2 - 2\operatorname{Im}(\sigma)\chi^2 + 4x\chi\chi')\psi^2\xi^{2s} + \text{terms belonging to } S^{2s-1} \quad (2.5.15)$$

Let

$$\alpha = (2s-1) + \operatorname{Im}(\sigma)$$

Then

$$q = (-\alpha\chi^2 + 4x\chi\chi')\psi^2|\xi|^{2s} + \text{terms belonging to } S^{2s-1} \quad (2.5.16)$$

By construction,

$$4x\chi\chi' \leq 0$$

Thus

$$-\alpha\chi^2 + 4x\chi\chi' \leq -\alpha\chi^2$$

We need $\alpha > 0$ or equivalently,

$$s > \frac{1}{2} - \operatorname{Im}(\sigma) \quad (2.5.17)$$

Under this condition,

$$-\alpha\chi^2 + 4x\chi\chi'$$

is strictly negative where $\chi \neq 0$. Thus,

$$q = \sigma_{2s}(Q) \leq -c_0\chi^2\psi^2|\xi|^{2s} + S^{2s-1} \quad (2.5.18)$$

Now let

$$B = Op(\chi(x)\psi(\xi)|\xi|^s)$$

So B belongs to Ψ^s and the principal symbol of B^*B is given by

$$|b|^2 = \chi^2\psi^2|\xi|^{2s}.$$

Now, choose

$$0 < c < c_0$$

and define

$$G := -Q - cB^*B$$

Then, from equation 2.5.18, we get

$$\begin{aligned} g &:= \sigma_{2s}(G) = -q - c|b|^2 \\ &\geq (c_0 - c)|b|^2 \\ &\geq 0 \end{aligned}$$

Thus by using the Sharp Garding Inequality 2.1.3 on the symbol g , we get

$$\operatorname{Re}\langle Op(g)u, u \rangle \geq -C\|u\|_{H^{s-\frac{1}{2}}}^2 \quad (2.5.19)$$

Thus

$$\operatorname{Re}\langle Gu, u \rangle \geq -C\|u\|_{H^{s-\frac{1}{2}}}^2 \quad (2.5.20)$$

Since

$$\langle Gu, u \rangle = -\langle Qu, u \rangle - c\langle B^*Bu, u \rangle$$

equation 2.5.20 implies

$$c\|Bu\|_{L^2}^2 \leq |\langle Qu, u \rangle| + C\|u\|_{H^{s-\frac{1}{2}}}^2 \quad (2.5.21)$$

Finally as

$$\langle Qu, u \rangle = 2\operatorname{Im}\langle P_\sigma u, Au \rangle,$$

we get

$$\|Bu\|_{L^2}^2 \lesssim |\langle P_\sigma u, Au \rangle| + \|u\|_{H^{s-\frac{1}{2}}}^2 \quad (2.5.22)$$

Now as $A \in \Psi^{2s-1}$,

$$A : H^s \rightarrow H^{1-s}$$

We know that

$$\|Au\|_{H^{1-s}} = \|\Lambda^{1-s}Au\|_{L^2} \quad \text{where } \Lambda = (1 + D_x^2)^{\frac{1}{2}}$$

So, by using properties of Theorem 2.1.1 and comparing symbols b and the symbol of $\Lambda^{1-s}A$, it is possible to arrive at the estimate

$$\|Au\|_{H^{1-s}} \lesssim \|Bu\|_{L^2} + \|u\|_{H^{s-\frac{1}{2}}} \quad (2.5.23)$$

Further, by using duality, we get

$$|\langle P_\sigma u, Au \rangle| \leq \|\tilde{\chi}P_\sigma u\|_{H^{s-1}} \|Au\|_{H^{1-s}} \quad \text{where } \tilde{\chi} \text{ is defined as in hypothesis}$$

This gives us

$$|\langle P_\sigma u, Au \rangle| \leq C\|\tilde{\chi}P_\sigma u\|_{H^{s-1}} \left(\|Bu\|_{L^2} + \|u\|_{H^{s-\frac{1}{2}}} \right) \quad (2.5.24)$$

Then by using Cauchy Schwartz with $\epsilon > 0$, we get

$$|\langle P_\sigma u, Au \rangle| \leq \epsilon\|Bu\|_{L^2}^2 + C_\epsilon\|\tilde{\chi}P_\sigma u\|_{H^{s-1}}^2 + C_\epsilon\|u\|_{H^{s-\frac{1}{2}}}^2 \quad (2.5.25)$$

Choosing $\epsilon > 0$ sufficiently small and absorbing the $\|Bu\|_{L^2}$ term to the left hand side, we can conclude

$$\|Bu\|_{L^2} \lesssim \|\tilde{\chi}P_\sigma u\|_{H^{s-1}} + \|\tilde{\chi}u\|_{H^{s-\frac{1}{2}}} \quad (2.5.26)$$

Now if we choose $\psi_0 \in C_0^\infty(\mathbb{R}_\xi)$ such that

$$1 = \psi_+(\xi) + \psi_-(\xi) + \psi_0(\xi)$$

where $\psi_+(\xi) = \psi(\xi)$ and $\psi_-(\xi) = \psi(-\xi)$, then

$$\chi u = Op(\chi\psi_+)u + Op(\chi\psi_-)u + Op(\chi\psi_0)u + \text{low order smoothing terms}$$

The first two terms are controlled by the two radial estimates. The last term is compactly supported in ξ and so it satisfies

$$\|Op(\chi\psi_0)u\|_{H^s} \lesssim \|u\|_{H^{s-\frac{1}{2}}} \quad (2.5.27)$$

Thus, using equations 2.5.26 and 2.5.27, we can conclude that

$$\|\chi u\|_{H^s} \leq C \left(\|\tilde{\chi} P_\sigma u\|_{H^{s-1}} + \|\tilde{\chi} u\|_{H^{s-\frac{1}{2}}} \right)$$

provided

$$s > \frac{1}{2} - \text{Im}(\sigma).$$

Analogous result to Proposition 2.5.2 for the adjoint operator on sinks

We now prove the corresponding radial estimate at the sink. In the sink case, one does not get an estimate from $P_\sigma^* u$ alone. An additional term is needed in a punctured neighbourhood of the radial set. This reflects the fact that bicharacteristics enter the radial set in forward time.

Proposition 2.5.3. *Let $s \in \mathbb{R}$ satisfy*

$$s < \frac{1}{2} + \text{Im}(\sigma).$$

Let $\chi \in C_0^\infty$ be supported sufficiently close to 0, with $\chi = 1$ near 0, and let $\psi \in C^\infty(\mathbb{R})$ localize to large negative ξ , so that $\psi(-\xi) = 1$ for $-\xi$ large. Then there exists an operator $G \in \Psi^s$, microsupported in a punctured neighbourhood of Σ_- , such that

$$\|\text{Op}(\psi(-\xi)\chi(x))u\|_{H^s} \lesssim \|P_\sigma^* u\|_{H^{s-1}} + \|Gu\|_{L^2} + \|u\|_{H^{s-\frac{1}{2}}}.$$

In the case of sinks, the inequality of the threshold gets reversed and we have an additional term on the right hand side of the estimate due to a difference in signs of the Hamiltonian calculation.

We are working on the set

$$\Sigma_- = \{x = 0, \xi < 0\}$$

Set

$$\mu = -\xi > 0$$

and let $\psi \in C^\infty(\mathbb{R})$ be defined as

$$\psi(\mu) = \begin{cases} 0 & \text{if } \mu \leq 1 \\ 1 & \text{if } \mu \geq 2 \end{cases}$$

Therefore $\psi(-\xi)$ localizes to large negative ξ .

Choose $\chi \in C_0^\infty((-\delta, \delta))$ such that

$$0 \leq \chi \leq 1, \quad \chi = 1 \quad \text{near } 0 \quad \text{and} \quad x\chi'(x) \leq 0$$

Moreover, choose χ so that $\text{supp } \chi'$ is contained in a punctured neighbourhood of 0.

Like in the case of source, choose

$$c(x, \xi) = \chi(x)\psi(\mu)\mu^{s-\frac{1}{2}} \quad (2.5.28)$$

and let

$$C = Op(c) \quad \text{and} \quad A = C^*C$$

A quick computation of the Hamiltonian gives us

$$H_p a = (2s - 1)\mu^{2s}\psi^2\chi^2 + 2\mu^{2s+1}\psi\psi'\chi^2 - 4x\mu^{2s}\psi^2\chi\chi' \quad (2.5.29)$$

The second term is a smoothing term supported in $1 \leq \mu \leq 2$ and can be ignored whereas the third term is positive. This is where the difference between a source and sink is noticed. In the case of a source, the third term was negative and of the same sign as the first, whereas in the case of a sink, this is not the case.

As earlier, we use the commutator identity 2.5.2 to get

$$2\operatorname{Im}\langle P_\sigma^* u, Au \rangle = \langle i[P_\sigma^*, A]u, u \rangle + 2\operatorname{Im}(\sigma)\langle D_x Au, u \rangle \quad (2.5.30)$$

and the principal symbol of

$$Q = i[P_\sigma^*, A] + i(P_\sigma - P_\sigma^*)A$$

is

$$H_p a + 2\operatorname{Im}(\sigma)\xi a = (2s - 1 - 2\operatorname{Im}(\sigma))\mu^{2s}\psi^2\chi^2 - 4x\mu^{2s}\psi^2\chi\chi' + \text{lower smoothing terms} \quad (2.5.31)$$

We now discuss the signs. Since we are proving the adjoint estimate at the sink, the estimate is valid in the range

$$s < \frac{1}{2} + \operatorname{Im}(\sigma).$$

Equivalently,

$$2s - 1 - 2\operatorname{Im}(\sigma) < 0.$$

Therefore the first term in equation 2.5.31 has negative sign. On the other hand, by our choice of χ , we have

$$x\chi'(x) \leq 0.$$

Since $\chi \geq 0$, this gives

$$-4x\chi\chi' \geq 0.$$

Thus the first term and the cutoff-derivative term in equation 2.5.31 have opposite signs. This is the main difference between the source estimate and the sink estimate. We need

$$2s - 1 - 2\operatorname{Im}(\sigma) < 0$$

or equivalently

$$s < \frac{1}{2} + \operatorname{Im}(\sigma)$$

Let

$$\gamma = 1 + 2\operatorname{Im}(\sigma) - 2s$$

and define

$$b_0(x, \xi) = \mu^s \psi(\mu) \chi(x) \quad \text{and} \quad B = \operatorname{Op}(b_0)$$

Thus

$$b_0^2 = \mu^{2s} \psi^2 \chi^2$$

and the main term is $-\gamma b_0^2$.

Let $\chi_1 \in C_0^\infty((-\delta, \delta))$ be such that

$$\chi_1 = 1 \quad \text{on a neighbourhood of } \operatorname{supp} \chi'.$$

Thus χ_1 is supported away from $x = 0$.

Now define

$$e(x, \xi) = 2\sqrt{-x\chi_1(x)\chi_1'(x)}\mu^s\psi(\mu)$$

Then

$$e^2 = -4x\chi_1\chi_1'\mu^{2s}\psi^2$$

and so

$$H_p a + 2 \operatorname{Im}(\sigma)\xi a = -\gamma b_0^2 + e^2 + \text{smoothing terms}$$

which after quantization gives us

$$i(P_\sigma A - AP_\sigma^*) = -\gamma B^* B + E^* E + R \quad \text{where } R \in \Psi^{2s-1} \quad (2.5.32)$$

Thus

$$\langle i(P_\sigma A - AP_\sigma^*)u, u \rangle \leq -\gamma \|Bu\|_{L^2}^2 + C\|Eu\|_{L^2}^2 + C\|u\|_{H^{s-\frac{1}{2}}}^2 \quad (2.5.33)$$

Since E is supported on χ' , choose $G \in \Psi^s$ elliptic on the wavefront set of E . This means that on a conic neighbourhood U of the $\operatorname{WF}'(E)$, we have

$$|g(x, \xi)| \geq c\langle \xi \rangle^s$$

for $|\xi|$ large. This gives us

$$\|Eu\|_{L^2} \lesssim \|Gu\|_{L^2} + \|u\|_{H^{s-\frac{1}{2}}} \quad (2.5.34)$$

Thus equations 2.5.33 and 2.5.34 give us

$$\langle i(P_\sigma A - AP_\sigma^*)u, u \rangle \leq -\gamma \|Bu\|_{L^2}^2 + C\|Gu\|_{L^2}^2 + C\|u\|_{H^{s-\frac{1}{2}}}^2$$

which gives us

$$\gamma \|Bu\|_{L^2}^2 \leq |2\operatorname{Im}\langle P_\sigma^* u, Au \rangle| + \|Gu\|_{L^2}^2 + \|u\|_{H^{s-\frac{1}{2}}}^2 \quad (2.5.35)$$

due to the identity

$$2\operatorname{Im}\langle P_\sigma^* u, Au \rangle = \langle i(P_\sigma A - AP_\sigma^*)u, u \rangle.$$

As in the case of source, since A has order $2s - 1$,

$$|\langle P_\sigma^* u, Au \rangle| \lesssim \|P_\sigma^* u\|_{H^{s-1}} \|Au\|_{H^{1-s}} \quad (2.5.36)$$

But as $A = C^* C$, through the same method as for the source estimate 2.5.23, we also have

$$\|Au\|_{H^{1-s}} \lesssim \|Bu\|_{L^2} + \|u\|_{H^{s-\frac{1}{2}}}$$

We conclude as in the proof of proposition 2.5.2.

Remark 2.5.4. *The positive commutator calculations above are initially only formal and require a regularization argument. The reason is that the commutant is constructed at the Sobolev order which the estimate is intended to prove. More precisely, we take $A = C^* C$ with $C \in \Psi^{s-\frac{1}{2}}$, so that $A \in \Psi^{2s-1}$, while the operator B controlled by the commutator belongs to Ψ^s . Thus $\|Bu\|_{L^2}$ measures precisely the microlocal H^s regularity of u , which is not known before the estimate has been proved.*

This creates a difficulty in the commutator identity

$$2\operatorname{Im}\langle P_\sigma u, Au \rangle = \langle (i[P_\sigma, A] + i(P_\sigma^* - P_\sigma)A)u, u \rangle.$$

For a smooth function u this identity follows directly by integration by parts. Under the hypotheses of the radial estimate, however, one only knows a priori that u has some lower Sobolev regularity $H^{s'}$, with $s' < s$. Since A has order $2s - 1$, this only gives

$$Au \in H^{s'-(2s-1)},$$

which need not be regular enough to pair with $P_\sigma u \in H^{s-1}$. Similarly, the expression

$$\langle B^* Bu, u \rangle = \|Bu\|_{L^2}^2$$

cannot be used before it is already known that $Bu \in L^2$. Without regularization, the argument would therefore use the desired H^s regularity in order to prove that same regularity.

To avoid this circular argument, one replaces the commutant by a family $A_t = C_t^* C_t$, $t > 0$, whose symbol contains an additional high frequency factor of the form

$$\phi_t(\mu) = (1 + t\mu)^{-(s-s')}.$$

Here, μ denotes the frequency variable. On the two radial components, it is given by $\mu = \xi$ or $\mu = -\xi$ respectively.

For each fixed $t > 0$, this factor lowers the order of C_t from $s - \frac{1}{2}$ to $s' - \frac{1}{2}$. Consequently, all pairings and quadratic forms are well-defined using only the a priori $H^{s'}$ regularity. At the same time,

$$\phi_t(\mu) \rightarrow 1 \text{ as } t \rightarrow 0,$$

so the regularized operators converge to the original commutant.

One must also check that differentiating ϕ_t in the Hamiltonian calculation does not destroy the sign of the commutator. The key bound is

$$0 \leq -\mu \frac{\phi_t'(\mu)}{\phi_t(\mu)} \leq s - s'.$$

At the source, the strict inequality placing s' above the radial threshold leaves enough positive margin to absorb this additional term. At the sink, the regularization term has the favourable sign, while the cutoff error remains supported in the punctured neighbourhood already controlled by the additional term in the sink estimate.

Finally, a smoothing family Λ_τ is inserted to justify rigorously the passing from

$$\langle P_\sigma u, A_t u \rangle - \langle A_t u, P_\sigma u \rangle$$

to the commutator pairing. After obtaining estimates uniform in t , one lets $t \rightarrow 0$ and uses weak compactness in L^2 to conclude that the unregularized quantity Bu belongs to L^2 .

For the complete regularization argument, including the treatment of the additional symbol term and the justification of the pairing identity, one can refer to section 5.4.7 given in *A Minicourse on Microlocal Analysis on Wave Propagation* by András Vasy [6].

2.6. Global Estimates

Proposition 2.6.1 ([10] Proposition 5.10). *Let $u \in \bar{H}^s$ and $s > \frac{1}{2} - \frac{1}{\kappa_h} \text{Im}(\sigma)$. Then we have*

$$\|u\|_{\bar{H}^s(X)} \leq C \left(\|\widehat{\square}_g(\sigma)u\|_{\bar{H}^{s-1}} + \|u\|_{\bar{H}^{s-\frac{1}{2}}(X)} \right) \quad (2.6.1)$$

Proof:

We choose $\widehat{\chi}_\pm$ as in proposition 2.5.1 and $\chi_\pm = 1$ on $(r \pm \frac{\delta}{2}, r \pm \frac{\delta}{2})$ and then choose $\widehat{\chi}_\pm$ such that

$$\text{supp} \widehat{\chi}_- \subset [r_- - \epsilon, r_- - \frac{\delta}{2}), \quad \text{supp} \widehat{\chi}_+ \subset [r_+ + \frac{\delta}{2}, r_+ + \epsilon)$$

and finally choose $\chi_0 \in C_0^\infty((r_- + \frac{\delta}{2}, r_+ - \frac{\delta}{2}))$ with the property that

$$1 = \widehat{\chi}_- + \widehat{\chi}_+ \chi_- + \chi_+ + \chi_0 \quad (2.6.2)$$

We can then apply proposition 2.5.1 to get the radial point estimates:

$$\|\chi_- u\|_{\bar{H}^s(X)} \lesssim \|\widehat{\square}_g(\sigma)u\|_{\bar{H}^{s-1}(X)} + \|u\|_{\bar{H}^{s-\frac{1}{2}}(X)} \quad (2.6.3)$$

and

$$\|\chi_+ u\|_{\bar{H}^s(X)} \lesssim \|\widehat{\square}_g(\sigma)u\|_{\bar{H}^{s-1}(X)} + \|u\|_{\bar{H}^{s-\frac{1}{2}}(X)} \quad (2.6.4)$$

Since $\text{supp}\chi_0 \subset (r_- + \frac{\delta}{2}, r_+ - \frac{\delta}{2})$, in this region $F > 0$ so we have elliptic type estimates like in proposition 2.4.1 and we have

$$\|\chi_0 u\|_{\bar{H}^s(X)} \lesssim \|\widehat{\square}_g(\sigma)u\|_{\bar{H}^{s-1}(X)} + \|u\|_{\bar{H}^{s-\frac{1}{2}}(X)} \quad (2.6.5)$$

Finally, $\widehat{\chi}_\pm$ are in the hyperbolic region where $F < 0$ and $\widehat{\square}_g(\sigma)$ is strictly hyperbolic. Thus using standard hyperbolic estimates, we get

$$\|\widehat{\chi}u\|_{\bar{H}^s(X)} \lesssim \|\widehat{\square}_g(\sigma)u\|_{\bar{H}^{s-1}(X)} + \|u\|_{\bar{H}^{s-\frac{1}{2}}(X)} \quad (2.6.6)$$

Thus combining equations 2.6.3, 2.6.4, 2.6.5 and 2.6.6, we get the global estimate on the entire manifold as

$$\|u\|_{\bar{H}^s(X)} \leq C \left(\|\widehat{\square}_g(\sigma)u\|_{\bar{H}^{s-1}} + \|u\|_{\bar{H}^{s-\frac{1}{2}}(X)} \right) \quad (2.6.7)$$

hence proving the proposition.

It is similarly possible to prove the following estimate for the adjoint:

$$\|v\|_{\dot{H}^{1-s}(X)} \leq C \left(\|\widehat{\square}_g^*(\sigma)v\|_{\dot{H}^{-s}} + \|v\|_{\dot{H}^{-s+\frac{1}{2}}(X)} \right) \quad (2.6.8)$$

2.7. Fredholm Property and Invertibility

We will now prove the Fredholm property and invertibility condition 1.1.3 of Theorem 1.1.1.

Let

$$X_0 = \bar{H}^{s+1}(X), \quad X_1 = \bar{H}^s(X) \quad \text{and} \quad X_2 = \bar{H}^{s-\frac{1}{2}}(X)$$

and

$$Y_0 = \bar{H}^{s-\frac{1}{2}}(X), \quad Y_1 = \bar{H}^{s-1}(X) \quad \text{and} \quad Y_2 = \bar{H}^{s-2}(X)$$

Then using the global estimates 2.6.7 and 2.6.8 from the previous section, the hypothesis for Lemma 2.3.1 is satisfied.

Since

$$\mathcal{A} = \{u \in X_1 : Pu \in Y_1\} = \{u \in \bar{H}^s(X) : \widehat{\square}_g(\sigma)u \in \bar{H}^{s-1}(X)\}$$

By lemma 2.3.1, the operator $\widehat{\square}_g(\sigma)$ is Fredholm from $\mathcal{X}^s$ to $\mathcal{Y}^{s-1}$.

Since we have shown that the operator $\widehat{\square}_g(\sigma)$ is Fredholm, invertibility is equivalent to showing

$$\ker P_\sigma = \{0\} \quad \text{and} \quad \text{coker } P_\sigma = \{0\}$$

where $P_\sigma = \widehat{\square}_g(\sigma)$.

We expand on the proof given by Häfner [10]. Now,

$$t_* = t + h(r)r_*(r) \quad (2.7.1)$$

where

$$h(r) = \begin{cases} 1 & \text{for } r \leq 3m \\ -1 & \text{for } r \geq 4m \end{cases} \quad (2.7.2)$$

Thus,

$$e^{-i\sigma t_*} u(r, \omega) = e^{-i\sigma t} (e^{-i\sigma h(r)r_*(r)} u(r, \omega)) \quad (2.7.3)$$

Let

$$v = e^{-i\sigma h(r)r_*(r)} u(r, \omega)$$

Now let us suppose by contradiction that for some non zero u ,

$$\widehat{\square}_g(\sigma)u = 0$$

By definition,

$$0 = \widehat{\square}_g(\sigma)u = e^{i\sigma t_*} \square_g(e^{-i\sigma t_*} u)$$

Therefore

$$\square_g(e^{-i\sigma t_*} u) = 0 \quad (2.7.4)$$

But

$$e^{-i\sigma t_*} u = e^{-i\sigma t} v$$

So we get

$$0 = \square_g(e^{-i\sigma t} v) = e^{-i\sigma t} \tilde{\square}_g(\sigma)v$$

where $\tilde{\square}_g(\sigma)$ is the Fourier transform with respect to t .

Therefore $\tilde{\square}_g(\sigma)v = 0$.

$$\begin{aligned} \implies \tilde{\square}_g(\sigma)e^{-i\sigma h(r)r_*(r)} u(r, \omega) &= 0 \\ \implies F\tilde{\square}_g(\sigma)e^{-i\sigma h(r)r_*(r)} u(r, \omega) &= 0 \end{aligned}$$

Also,

$$F\tilde{\square}_g(\sigma) = r^{-2} F D_r (r^2 F) D_r - r^{-2} F \Delta_{\mathbb{S}^2} - \sigma^2$$

Note that

$$F\tilde{\square}_g(\sigma) = F\tilde{\square}_g(0) - \sigma^2$$

and so

$$(F\tilde{\square}_g(0) - \sigma^2) v = 0.$$

Thus v is an L^2 eigenfunction of $\tilde{\square}_g(0)$.

We now show that $F\tilde{\square}_g(0)$ is a self adjoint positive operator on the Hilbert space

$$\mathcal{H} = L^2((r_-, r_+) \times \mathbb{S}^2; F^{-1} r^2 dr d\omega).$$

To obtain a self adjoint operator on $\mathcal{H}$, one must specify a domain. We first define the minimal operator

$$A_0 w := r^{-2} F D_r (r^2 F) D_r w - r^{-2} F \Delta_{\mathbb{S}^2} w$$

on the dense domain

$$\mathcal{D}(A_0) = C_c^\infty((r_-, r_+) \times \mathbb{S}^2).$$

Thus

$$A_0 : C_c^\infty((r_-, r_+) \times \mathbb{S}^2) \subset \mathcal{H} \longrightarrow \mathcal{H}.$$

We shall show that A_0 is densely defined, symmetric, and nonnegative. Then the Friedrichs extension theorem gives a canonical self adjoint nonnegative extension. The Hilbert space inner product on $\mathcal{H}$ is

$$\langle u, v \rangle_{\mathcal{H}} = \int_{r_-}^{r_+} \int_{\mathbb{S}^2} u(r, \omega) \overline{v(r, \omega)} F^{-1} r^2 dr d\omega.$$

Let

$$\phi, \psi \in C_c^\infty((r_-, r_+) \times \mathbb{S}^2).$$

We compute

$$\langle A_0 \phi, \psi \rangle_{\mathcal{H}}.$$

By definition,

$$A_0 \phi = r^{-2} F D_r (r^2 F) D_r \phi - r^{-2} F \Delta_{\mathbb{S}^2} \phi.$$

Hence

$$\begin{aligned} \langle A_0 \phi, \psi \rangle_{\mathcal{H}} &= \int_{r_-}^{r_+} \int_{\mathbb{S}^2} (r^{-2} F D_r (r^2 F) D_r \phi - r^{-2} F \Delta_{\mathbb{S}^2} \phi) \bar{\psi} F^{-1} r^2 dr d\omega \\ &= \int D_r (r^2 F D_r \phi) \bar{\psi} dr d\omega - \int \Delta_{\mathbb{S}^2} \phi \bar{\psi} dr d\omega. \end{aligned} \quad (2.7.5)$$

Consider the radial term

$$I_r = \int D_r (r^2 F D_r \phi) \bar{\psi} dr d\omega.$$

Since

$$D_r = -i \partial_r,$$

we have

$$I_r = \int -i \partial_r (r^2 F D_r \phi) \bar{\psi} dr d\omega.$$

Integrating by parts in r , and using the compact support of ϕ and ψ in the open interval (r_-, r_+) , there is no boundary term. Thus

$$I_r = \int i (r^2 F D_r \phi) \partial_r \bar{\psi} dr d\omega. \quad (2.7.6)$$

Now

$$D_r \psi = -i \partial_r \psi.$$

Therefore

$$\overline{D_r \psi} = \overline{-i \partial_r \psi} = i \partial_r \bar{\psi}.$$

Hence

$$i \partial_r \bar{\psi} = \overline{D_r \psi}.$$

Substituting this into expression 2.7.6 gives

$$I_r = \int r^2 F D_r \phi \overline{D_r \psi} dr d\omega.$$

Now consider the angular term

$$I_\omega = - \int \Delta_{\mathbb{S}^2} \phi \bar{\psi} dr d\omega.$$

For each fixed r , we use integration by parts on the closed manifold $\mathbb{S}^2$. Since $\mathbb{S}^2$ has no boundary,

$$- \int_{\mathbb{S}^2} \Delta_{\mathbb{S}^2} \phi \bar{\psi} d\omega = \int_{\mathbb{S}^2} \langle \nabla_\omega \phi, \nabla_\omega \psi \rangle_\omega d\omega.$$

Therefore

$$I_\omega = \int_{r_-}^{r_+} \int_{\mathbb{S}^2} \langle \nabla_\omega \phi, \nabla_\omega \psi \rangle_\omega d\omega dr. \quad (2.7.7)$$

Here ∇_ω denotes the gradient with respect to the round metric on $\mathbb{S}^2$, and $\langle \cdot, \cdot \rangle_\omega$ denotes the pointwise inner product induced by that metric.

More explicitly, if

$$d\omega^2 = d\theta^2 + \sin^2 \theta d\varphi^2,$$

then

$$d\omega = \sin \theta d\theta d\varphi,$$

and

$$\langle \nabla_\omega \phi, \nabla_\omega \psi \rangle_\omega = \partial_\theta \phi \overline{\partial_\theta \psi} + \frac{1}{\sin^2 \theta} \partial_\varphi \phi \overline{\partial_\varphi \psi}.$$

With the convention

$$\Delta_{\mathbb{S}^2} = \frac{1}{\sin \theta} \partial_\theta (\sin \theta \partial_\theta) + \frac{1}{\sin^2 \theta} \partial_\varphi^2,$$

one has

$$\int_{\mathbb{S}^2} \Delta_{\mathbb{S}^2} \phi \overline{\psi} d\omega = - \int_{\mathbb{S}^2} \langle \nabla_\omega \phi, \nabla_\omega \psi \rangle_\omega d\omega.$$

Combining the radial and angular computations, we obtain

$$\langle A_0 \phi, \psi \rangle_{\mathcal{H}} = \int r^2 F D_r \phi \overline{D_r \psi} dr d\omega + \int \langle \nabla_\omega \phi, \nabla_\omega \psi \rangle_\omega dr d\omega. \quad (2.7.8)$$

This identity is the main computation.

We now show that A_0 is symmetric.

Starting from the identity above,

$$\langle A_0 \phi, \psi \rangle_{\mathcal{H}} = \int r^2 F D_r \phi \overline{D_r \psi} dr d\omega + \int \langle \nabla_\omega \phi, \nabla_\omega \psi \rangle_\omega dr d\omega.$$

Similarly,

$$\langle A_0 \psi, \phi \rangle_{\mathcal{H}} = \int r^2 F D_r \psi \overline{D_r \phi} dr d\omega + \int \langle \nabla_\omega \psi, \nabla_\omega \phi \rangle_\omega dr d\omega.$$

Taking complex conjugates gives

$$\overline{\langle A_0 \psi, \phi \rangle_{\mathcal{H}}} = \int r^2 F D_r \phi \overline{D_r \psi} dr d\omega + \int \langle \nabla_\omega \phi, \nabla_\omega \psi \rangle_\omega dr d\omega.$$

Therefore

$$\langle A_0 \phi, \psi \rangle_{\mathcal{H}} = \overline{\langle A_0 \psi, \phi \rangle_{\mathcal{H}}}. \quad (2.7.9)$$

Equivalently,

$$\langle A_0 \phi, \psi \rangle_{\mathcal{H}} = \langle \phi, A_0 \psi \rangle_{\mathcal{H}}.$$

Hence A_0 is symmetric on $C_c^\infty((r_-, r_+) \times \mathbb{S}^2)$.

Now set $\psi = \phi = w$. Then the identity becomes

$$\langle A_0 w, w \rangle_{\mathcal{H}} = \int r^2 F |D_r w|^2 dr d\omega + \int |\nabla_\omega w|_\omega^2 dr d\omega.$$

Inside the region $r_- < r < r_+$,

$$F(r) > 0$$

Therefore

$$r^2 F |D_r w|^2 \geq 0$$

and

$$|\nabla_\omega w|_\omega^2 \geq 0.$$

Hence

$$\langle A_0 w, w \rangle_{\mathcal{H}} \geq 0.$$

Thus A_0 is nonnegative. In particular, it is bounded below with lower bound 0:

$$A_0 \geq 0.$$

This means

$$\langle A_0 w, w \rangle_{\mathcal{H}} \geq 0 \cdot \|w\|_{\mathcal{H}}^2$$

for every

$$w \in C_c^\infty((r_-, r_+) \times \mathbb{S}^2).$$

The domain

$$C_c^\infty((r_-, r_+) \times \mathbb{S}^2)$$

is dense in

$$L^2((r_-, r_+) \times \mathbb{S}^2; F^{-1} r^2 dr d\omega).$$

Therefore A_0 is densely defined and we have also shown that A_0 is symmetric and nonnegative. The Friedrichs extension theorem says that if a densely defined symmetric operator A_0 on a Hilbert space is bounded below, then it admits a canonical self-adjoint extension A_F , called its Friedrichs extension. Moreover, the Friedrichs extension has the same lower bound. Since we are in the same situation, therefore the Friedrichs extension theorem gives a self-adjoint operator

$$A_F : \mathcal{D}(A_F) \subset \mathcal{H} \rightarrow \mathcal{H}$$

such that

$$A_F \supset A_0$$

and

$$A_F \geq 0.$$

Thus the Friedrichs realization of

$$F\tilde{\square}_g(0) = r^{-2} F D_r (r^2 F) D_r - r^{-2} F \Delta_{\mathbb{S}^2}$$

is self-adjoint and positive on

$$L^2((r_-, r_+) \times \mathbb{S}^2; F^{-1} r^2 dr d\omega).$$

Now suppose

$$\sigma = a + ib \quad \text{with } b > 0$$

$$\sigma^2 = a^2 - b^2 + 2iab$$

If $a \neq 0$, then σ^2 is not real and so it cannot be the eigenvalue of a self adjoint operator.

If $a = 0$, then

$$\sigma^2 = -b^2 < 0$$

Thus it cannot be the eigenvalue of a positive operator.

So we conclude that $v = 0$ or equivalently,

$$u = 0$$

on $(r_-, r_+) \times \mathbb{S}^2$. Thus our assumption was incorrect and

$$\ker P_\sigma = \{0\}.$$

This can be extended to the space

$$(r_- - \epsilon, r_+ + \epsilon) \times \mathbb{S}^2.$$

For details, one can refer to the paper by Zworski [14].

It remains to prove that

$$\text{coker } P_\sigma = \{0\}.$$

Since P_σ is Fredholm, this is equivalent to showing

$$\ker P_\sigma^* = \{0\}.$$

Let

$$v \in \ker P_\sigma^* \subset (Y^{s-1})^* = \dot{H}^{1-s}(X).$$

The point which requires additional care, compared with the proof of $\ker P_\sigma = \{0\}$, is the choice of the local solution at the horizons. The condition $v \in \dot{H}^{1-s}(X)$ does not by itself exclude the non-smooth branch. Instead, the correct branch is selected by combining the support property of v , the hyperbolic estimates outside (r_-, r_+) , and the equation $P_\sigma^* v = 0$.

Since $v \in \dot{H}^{1-s}(X)$, its support is contained in the extended interval X , and hence v vanishes near the two endpoints of X . Moreover, P_σ^* is hyperbolic in the regions

$$r_- - \varepsilon_M < r < r_- \quad \text{and} \quad r_+ < r < r_+ + \varepsilon_M.$$

Uniqueness for the homogeneous equation $P_\sigma^* v = 0$ therefore gives

$$v = 0 \quad \text{for } r < r_- \quad \text{and} \quad v = 0 \quad \text{for } r > r_+.$$

Consequently,

$$\text{supp } v \subset [r_-, r_+] \times \mathbb{S}^2.$$

The mechanism can be understood from the model operator

$$P_\sigma^* = D_x x D_x - \bar{\sigma} D_x.$$

Suppose that v is supported in $\{x \geq 0\}$ and satisfies $P_\sigma^* v = 0$. On $x > 0$, the two solutions are 1 and $x^{i\bar{\sigma}}$, so near $x = 0$ one may write

$$v = C_1 H(x) + C_2 x_+^{i\bar{\sigma}} + \sum_{j=0}^N a_j \delta_0^{(j)}.$$

where H is the Heaviside function and $x_+^{i\bar{\sigma}} = H(x) x^{i\bar{\sigma}}$.

A direct computation gives

$$P_\sigma^* H = i\bar{\sigma} \delta_0, \quad P_\sigma^* x_+^{i\bar{\sigma}} = 0, \quad P_\sigma^* \delta_0^{(j)} = (j + 1 + i\bar{\sigma}) \delta_0^{(j+1)}.$$

The coefficient of $\delta_0^{(N+1)}$ must vanish, and hence

$$a_N (N + 1 + i\bar{\sigma}) = 0.$$

Since

$$\text{Re}(N + 1 + i\bar{\sigma}) = N + 1 + \text{Im } \sigma > 0,$$

we have $N + 1 + i\bar{\sigma} \neq 0$, and therefore $a_N = 0$. Repeating the argument gives

$$a_0 = \cdots = a_N = 0,$$

and the remaining term $i\bar{\sigma}C_1\delta_0$ then gives $C_1 = 0$.

Hence the support condition and the equation select the solution

$$v = C_2 x_+^{i\bar{\sigma}},$$

rather than the constant solution.

The same idea applies for our operator near $r = r_-$ and $r = r_+$. After separating the angular variables, the two local solutions of the adjoint equation near $r = r_-$ are of the form

$$f_-(r) \quad \text{and} \quad (r - r_-)^{i\bar{\sigma}/\kappa_-} g_-(r),$$

where f_- and g_- are smooth and nonzero at $r = r_-$. Since v vanishes for $r < r_-$, each angular component can be written locally in the form

$$\mathbf{1}_{(r_-, r_+)} \left(C_1 f_-(r) + C_2 (r - r_-)^{i\bar{\sigma}/\kappa_-} g_-(r) \right) + \sum_{j=0}^N a_j \delta_{r_-}^{(j)}.$$

As in the model problem, applying P_σ^* to this expression forces

$$C_1 = 0 \quad \text{and} \quad \sum_{j=0}^N a_j \delta_{r_-}^{(j)} = 0.$$

Thus, near $r = r_-$, the selected solution has the form

$$C_2 \mathbf{1}_{(r_-, r_+)} (r - r_-)^{i\bar{\sigma}/\kappa_-} g_-(r).$$

We can make a similar argument near $r = r_+$.

Therefore, at both horizons, it is the non-smooth branch which is selected by the support condition and the equation $P_\sigma^* v = 0$. For the full details of this argument, one can refer to [12, Proposition 6.40].

Since $\text{Im } \sigma > 0$, we have

$$\text{Re} \left(\frac{i\bar{\sigma}}{\kappa_h} \right) = \frac{\text{Im } \sigma}{\kappa_h} > 0.$$

Thus the selected solution vanishes at the horizon, even though it does not necessarily extend smoothly across it. This vanishing compensates for the growth produced by the change from t_* to t .

Indeed, set

$$w = e^{-i\bar{\sigma}h(r)r_*(r)} v.$$

Near $r = r_-$, with $x = r - r_-$, one has

$$r_*(r) = \frac{1}{2\kappa_-} \log x + O(1)$$

and the selected solution satisfies

$$v \sim x^{i\bar{\sigma}/\kappa_-}.$$

Consequently,

$$w \sim x^{-i\bar{\sigma}/(2\kappa_-)} x^{i\bar{\sigma}/\kappa_-} = x^{i\bar{\sigma}/(2\kappa_-)}.$$

Since $F(r) \sim 2\kappa_- x$, we obtain

$$|w|^2 F^{-1} dr \sim x^{-1+\text{Im } \sigma/\kappa_-} dx,$$

which is locally integrable because $\text{Im } \sigma > 0$. The same calculation applies near $r = r_+$. Hence

$$w \in L^2 \left((r_-, r_+) \times \mathbb{S}^2; F^{-1} r^2 dr d\omega \right).$$

The same asymptotic behavior shows that w satisfies the Friedrichs conditions and belongs to the domain of the Friedrichs realization of $F\tilde{\square}_g(0)$.

Moreover,

$$\left(F\tilde{\square}_g(0) - \bar{\sigma}^2\right)w = 0.$$

But $F\tilde{\square}_g(0)$ is self-adjoint and nonnegative on this Hilbert space, while

$$\bar{\sigma}^2 \notin [0, \infty)$$

when $\text{Im } \sigma > 0$. Hence $w = 0$, and therefore $v = 0$. Thus

$$\ker P_\sigma^* = \{0\},$$

so

$$\text{coker } P_\sigma = \{0\}.$$

Thus we have shown that P_σ is bijective for $\text{Im}(\sigma) > 0$ and since P_σ is Fredholm, bijectivity implies the existence of a bounded inverse by the open mapping theorem. Thus

$$P_\sigma^{-1} : \mathcal{Y}^{s-1} \rightarrow \mathcal{X}^s$$

exists for $\text{Im}(\sigma) > 0$.

Finally, for fixed $s \geq \frac{1}{2}$, set

$$\Omega_s = \left\{ \sigma \in \mathbb{C} : \text{Im } \sigma > \frac{1-2s}{2\gamma} \right\}.$$

This is a connected open subset of $\mathbb{C}$. Moreover, since

$$\frac{1-2s}{2\gamma} \leq 0,$$

we have

$$\{\text{Im}(\sigma) > 0\} \subset \Omega_s$$

and since P_σ is invertible there, its Fredholm index is zero at those points.

The map $\sigma \mapsto P_\sigma$ is an analytic family of Fredholm operators on Ω_s , and the Fredholm index is locally constant. Since Ω_s is connected and contains points with $\text{Im}(\sigma) > 0$, it follows that

$$\text{ind } P_\sigma = 0 \quad \text{for every } \sigma \in \Omega_s.$$

Remark 2.7.1. *The condition $\text{Im}(\sigma) > 0$ in the invertibility statement must be strict. Indeed, the constant function gives a non-zero element of the kernel at $\sigma = 0$.*

Recall that

$$\hat{\square}_g(\sigma)u = e^{i\sigma t_*} \square_g(e^{-i\sigma t_*} u).$$

Setting $\sigma = 0$, we obtain

$$\hat{\square}_g(0)u = \square_g u.$$

Since the wave operator is given by

$$\square_g u = \frac{1}{\sqrt{|\det g|}} \partial_\mu \left(\sqrt{|\det g|} g^{\mu\nu} \partial_\nu u \right),$$

it follows immediately that

$$\square_g 1 = 0,$$

because all derivatives of the constant function vanish. Therefore

$$\widehat{\square}_g(0)1 = 0.$$

Since 1 is smooth and non-zero, we have

$$1 \in \ker \widehat{\square}_g(0),$$

and hence

$$\ker \widehat{\square}_g(0) \neq \{0\}.$$

Thus $\widehat{\square}_g(0)$ is not injective and therefore is not invertible.

Consequently, the invertibility conclusion cannot be extended from $\text{Im}(\sigma) > 0$ to $\text{Im}(\sigma) \geq 0$.

3. Broader Context

The theorem proved in the previous section is not only a result about one particular wave operator. It is also a simple model for a much broader method used in the study of waves on black hole spacetimes. The Schwarzschild-de Sitter case is especially useful because its geometry is simple enough that many calculations can be written explicitly, while it already contains the main features which appear in the Kerr-de Sitter problem: horizons, radial points, trapping and resonances.

The purpose of this section is to step back from the details of the proof and explain three questions. First, which parts of the argument are determined by the principal part of the operator, and which parts may change when lower order terms are added? Second, how does the theorem fit into the more general study of linear waves on Kerr-de Sitter spacetime? Finally, why is the Fredholm theorem only one part of the analysis, and why are additional high frequency estimates needed in order to obtain decay results?

3.1. Which parts depend only on the principal part?

The principal symbol of the stationary operator considered in this thesis is

$$p(r, \omega, \xi, \eta) = F(r)\xi^2 + r^{-2}|\eta|^2.$$

It describes the highest-order part of $\widehat{\square}_g(\sigma)$. Terms containing fewer derivatives do not enter the principal symbol.

The principal symbol determines the basic geometric picture used throughout the proof. It determines where the operator is elliptic, where it is characteristic, and how singularities propagate. More precisely, it determines the characteristic set

$$\Sigma = \{p = 0\},$$

the Hamiltonian vector field H_p , and the corresponding bicharacteristic flow. It therefore also determines the location of the radial sets and whether they behave as sources or sinks for the Hamiltonian flow.

Suppose now that the operator is replaced by

$$\widehat{\square}_g(\sigma) + L(\sigma),$$

where $L(\sigma)$ is a lower order operator. Since the principal symbol is unchanged, the elliptic region, the characteristic set, the bicharacteristic flow and the trapped set remain the same. Thus the broad geometric structure of the microlocal argument is unchanged.

This does not mean that the entire theorem remains unchanged. In particular, the precise Sobolev threshold in the radial point estimate is not determined by the principal symbol alone. Recall that the radial estimate in the present problem requires

$$s > \frac{1}{2} - \frac{\text{Im}(\sigma)}{\kappa_{\pm}}.$$

Imposing this regularity excludes the non-smooth branch at the corresponding horizon.

The appearance of $\text{Im}(\sigma)$ in this inequality comes from a lower order part of the stationary operator. This can already be seen in the model operator

$$P_{\sigma} = D_x x D_x - \sigma D_x.$$

Its principal symbol is

$$p = x\xi^2,$$

which determines the radial source and sink structure. However, the term $-\sigma D_x$ is of lower order and enters the positive commutator calculation. It is this term which produces the threshold

$$s > \frac{1}{2} - \text{Im}(\sigma).$$

Thus the principal symbol determines where the radial points are and how the Hamiltonian flow behaves near them, while the lower order terms help determine how much Sobolev regularity is required there.

Consequently, adding a general lower order term can change the radial point threshold. The condition

$$\text{Im}(\sigma) > \frac{1 - 2s}{2\gamma}$$

which appears in the Fredholm theorem may therefore have to be replaced by a different condition. The general shape of the proof remains the same, but the precise half plane in which the operator is Fredholm can shift.

The effect depends on the type of lower order term. A first-order term can contribute directly to the radial threshold. A smooth zero order potential does not normally enter the same leading radial calculation, but it can still change the kernel, the resonances and the invertibility properties of the operator.

We may summarize the distinction as follows. The principal symbol determines the paths along which waves and singularities travel. The lower order terms do not change these paths, but they can change the regularity conditions at the horizons and the frequencies at which non zero homogeneous solutions exist.

3.2. Relation with the Kerr–de Sitter problem

The Schwarzschild-de Sitter metric is the non rotating member of the Kerr–de Sitter family. Its spherical symmetry makes the wave equation particularly easier to work with. One can expand a solution in spherical harmonics,

$$u(r, \omega) = \sum_{\ell=0}^{\infty} \sum_{m=-\ell}^{\ell} R_{\ell m}(r) Y_{\ell m}(\omega),$$

and reduce the stationary equation to a family of ordinary differential equations.

This simplification is useful for understanding individual mode solutions, the behaviour near the horizons and the zero resonance. However, the importance of the Fredholm argument is that it is not fundamentally based on the spherical harmonic decomposition. Its main ingredients are

instead the geometry of the principal symbol, propagation of regularity, radial point estimates and compactness.

This becomes important when passing to Kerr-de Sitter spacetime. The rotating spacetime is no longer spherically symmetric, and the stationary wave operator is more complicated. Nevertheless, the main geometric features remain similar. There is still an event horizon and a cosmological horizon, and the Hamiltonian flow still has radial behaviour near these horizons. The trapping also remains normally hyperbolic.

For this reason, the analogue of the Fredholm theorem for Kerr-de Sitter follows the same general plan as in the Schwarzschild-de Sitter case. One proves elliptic estimates away from the characteristic set, propagation estimates in the ordinary characteristic region, and radial point estimates near the horizons. These estimates are then combined into a global estimate with a compact error term, from which the Fredholm property follows.

Rotation nevertheless introduces new difficulties. One of them is related to the absence of a globally timelike stationary Killing vector field. As a result, the simple positive self adjoint argument used in the Schwarzschild-de Sitter invertibility proof does not pass directly to Kerr-de Sitter.

The Fredholm framework separates this difficulty from the rest of the analysis. The microlocal estimates still give a Fredholm family. The remaining question is then whether there exist non zero mode solutions at particular values of σ . In this way, the geometric part of the proof and the mode analysis can be treated separately.

The theorem proved in this thesis should therefore be viewed as the simplest example of the framework used for Kerr-de Sitter. It exhibits the main structure without the additional complications caused by rotation.

3.3. Why high frequency estimates are also needed

The theorem proved above is essentially a bounded-frequency result. It shows that, for each fixed σ in the appropriate region,

$$\widehat{\square}_g(\sigma) : \mathcal{X}^s \longrightarrow \mathcal{Y}^{s-1}$$

is Fredholm. It also gives invertibility in the upper half plane and, through analytic Fredholm theory, a meromorphic inverse in the Fredholm region.

This is enough to define resonances as poles of

$$\widehat{\square}_g(\sigma)^{-1}.$$

It is not, however, enough to prove a decay statement for solutions of the time-dependent wave equation.

To understand why, recall that the solution is studied by taking the Fourier transform in t_* . Formally, it can be recovered by an inverse transform of the form

$$u(t_*) = \frac{1}{2\pi} \int_{\text{Im}(\sigma)=C} e^{-i\sigma t_*} \widehat{u}(\sigma) d\sigma,$$

where the line $\text{Im}(\sigma) = C$ is initially chosen in the region of invertibility.

In order to obtain decay, one would like to move this contour into the lower half plane. The poles crossed during this deformation give the resonant terms in the asymptotic expansion. The remaining integral gives the decaying remainder.

The Fredholm theorem tells us that the poles are isolated and have finite multiplicity. It does not give uniform control of the inverse when

$$|\text{Re}(\sigma)| \longrightarrow \infty.$$

Without such control, the contour deformation and the estimate of the remaining integral cannot be justified. This is why an additional high frequency analysis is needed.

The complete analysis may be divided into three parts. First, the radial point and global estimates give a Fredholm family and hence a meromorphic inverse. Second, the low frequency or mode analysis determines which resonances lie close to the real axis. In the scalar Schwarzschild–de Sitter problem, $\sigma = 0$ is a resonance corresponding to the constant solution. Third, the high frequency estimates control the inverse for large $|\operatorname{Re}(\sigma)|$ and provide a resonance free region apart from the resonances already detected at bounded frequency.

Combining these ingredients gives an asymptotic expansion of the solution in terms of resonances. Since the zero resonance is present, a general solution does not decay to zero.

Thus the result proved in this thesis is the first major step in the spectral analysis. It provides the Fredholm setting in which resonances can be defined and studied. The additional high frequency estimates are what turn this spectral framework into quantitative statements about the long time behaviour of waves.

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